



SMART DMV - Digital Verification System

Group 4 : Enoch Eugene Saka, Flora Phiri, Samuel Akuffo, Innocent Kuda Tsenzete , Srinivas

Vengaldas, Zhenyu Zhu

The George Washington University, School of Business

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Dr. Soheil Sibdari

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Abstract

By relying on manual, in-person verification of documents submitted to the DMV, the traditional DMV process creates operational bottlenecks, high levels of customer service rejections and long wait times. To alleviate these issues, this capstone project proposes the Smart-DMV Digital Verification System : Developed using modern technologies including Next.js and Supabase, the Smart-DMV Digital Verification System utilizes an Agentic AI pipeline through which Google Vision will be used for high fidelity text extraction from documents submitted to the DMV and the Gemini API will be used for more advanced reasoning capabilities.

The Smart-DMV Digital Verification System meets the government's strict security requirements by utilizing a confidence score between 90% - 100% to trigger an automated approval for a resident's appointment to be scheduled. To address complex edge cases where there is a mismatch in data between what was submitted by the resident and what is on file with the government, the Smart-DMV Digital Verification System utilises Human in the Loop through the use of a Generative AI Staff Co-Pilot which provides structured summaries of mismatches from manual reviews to achieve transparency and explainable decision-making.

According to operational analysis, the digital-first approach is projected to achieve a 70-80% automation rate, decrease the percentage of unprepared visits from 20% to 5%, and reduce the cost to process each transaction by 51%. The projected return on investment (ROI) for Year 1 will be 198% as a result of maximising capacity without physical infrastructure expansion. Finally, the Smart-DMV provides a secure and scalable digital transformation roadmap across public sectors, showcasing how Agentic AI enables modernisation of government service deliveries with strict regulatory compliance and accountability.

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1. Introduction

The D.C. DMV performs driving licenses, vehicle registrations, and identity verifications for D.C. That means that the DMV works with approximately 3200 residents every day and 650,000 licensed drivers throughout D.C. There are more residents and drivers using the DMV than almost any other agency that reports to the Office of the City Administrator(Office of the City Administrator, 2024).

In FY2024, the DMV issued identity documents to 77,271 residents and processed more than 392,000 online transactions over the Internet. Given the volume of activity, the DMV has a very complex interaction environment with high consequence, multiple layers of regulatory compliance, and trust from the public, reliability of service and, importantly, supporting the safe and secure functioning of Government and D.C (DC DMV, n.d.).

The DMV, nevertheless, does experience significant challenges with its operation (manual document verification, inconsistent appointment preparation, high customer volumes creating excessive pressure on the existing workforce) resulting in long wait times, return visits, and reduced productivity of residents and DMV employees.

1.1. System Introduction

This capstone project presents the Smart-DMV, a digital verification platform that uses Artificial Intelligence to support a regime of ongoing operational improvement. The Smart-DMV will enhance the support of pre-appointment workflow processes, automate the intake of documents and improve the amount of cases processed by government Services. The Smart-DMV supports a modern, multiuser web-based environment in which citizens can upload their identification and residency documents. The documents are immediately evaluated by the Google Vision API and, as a backup, the Paddle OCR model, to convert unstructured text into structured

data. The Smart-DMV has a robust core logic component that utilizes a dynamic confidence scoring evaluation process to evaluate every document submitted to the system. Any documents with an exact match rating of 90 - 100% will be automatically approved to ensure quick case processing. Any documents that do not have an exact match or that have only minor discrepancies (i.e. 60 - 89%) will be compiled into a Review Package that includes AI-generated rationales and forwarded to government Services staff for review and manual exception processing.

To stop any bottlenecks with appointment scheduling, the process is gated so that only those users with verified status can schedule one of the limited slots (either physically or virtually). Additionally, the system maintains strong security and compliance by masking PII of any document that is stored in a secure cloud in order to protect any sensitive information from being viewed by unauthorized parties (and to comply with regulations). Exception reporting and analytics for both performance-related data and operational statistics are also made available for review via the web or our client application. Through a combination of automating routine document ingestion and implementing a human-in-the-loop solution for intelligent workflow, the Smart-DMV system will dramatically decrease the amount of operational activities that create inefficiencies while also providing a much-improved resident experience.

1.2. Project Charter

The Project Charter (*see Table 1 in Appendix*) formally authorizes the DC DMV Digital Document Verification System, defining the strategic mandate to replace manual processing with an AI-driven model that targets wait times under 15 minutes. It establishes the project scope and critical success metrics, while explicitly empowering the project team to utilize academic resources for development. This document aligns stakeholders on the timeline and deliverables, serving as

the definitive approval to proceed with the system's design and implementation between February and April 2026.

2. Business Case

2.1. Organization Description

Functioning within a complex regulatory environment, the DMV in D.C. has a lot of work to do. First, they have to follow the law on federal levels and in D.C. When following the rules about identity, REAL ID, and safety, the DMV also needs to work closely with law enforcement (OCTO), the department of transportation, and also national systems that verify or validate identity like Social Security (AAMVA). The complexities of these intertwined “eco-systems” put a lot of pressure on the DMV to maintain accurate information, process large volumes of sensitive personal data, and provide services consistently, transparently, and reliably.

In the past few years, the DMV has worked towards creating more digital services in order to meet customer demands and to relieve congestion from in-person services. With more than 650,000 licensed drivers and the thousands of daily transactions performed, the DMV has started using more technology in order to make processes easier, provide more accurate identity verification, and become more efficient in delivering services. However, the DMV’s ability to provide good, timely service is still being impacted by the use of manual document reviews, appointment bottlenecks, and customer pre-visit deficiencies. All of these operational pressures demonstrate a continued need for modernizing services to enhance service delivery, decrease administrative workloads, and support long-term strategic initiatives such as Smart DMV Digital Verification.

2.1.1. Environment

The current operating environment relies heavily on in-person transactions and manual document verification, leading to significant service bottlenecks (District of Columbia Department of Motor Vehicles [DC DMV], 2025).

Internal audits reveal that approximately 30–35% of customers arrive with incomplete or incorrect documentation, resulting in prolonged wait times often exceeding 45 minutes and requiring multiple visits (DC DMV, 2025). This manual, walk-in model not only diminishes customer satisfaction but also diverts staff from higher-value tasks, with an estimated 25–30% of staff time dedicated to repetitive verification duties (DC DMV, 2025).

Furthermore, the agency faces a "Silver Tsunami" of retiring veteran staff coupled with a surge in complex REAL ID transactions, creating a standard industry challenge that the DC DMV must navigate (American Association of Motor Vehicle Administrators [AAMVA], 2024).

Technologically, existing systems offer limited pre-visit validation or real-time feedback, creating an inefficient and frustrating experience for both citizens and employees (DC DMV, 2025).

2.1.2. Business Goals

DC DMV's Digital Document Verification System implements an AI-based digital solution to promote mainframe automation of outdated, legacy processes (Washington D.C. Department of Motor Vehicles, 2025). This solution allows the agency to pre-verify customer documentation prior to making an appointment, thus moving document verification up-stream as a digital clerk and alleviating physical congestion while also dramatically improving the accuracy of initial submissions.

The following goals were established to align with public sector digital transformation initiatives focused on enhancing operational efficiency and building civic trust, Key business goals and associated metrics are outlined in table 2 (*see Table 2 in Appendix*):

- I. **Eliminating Operational Bottlenecks:** Reduce the average time a customer waits in line to less than 15 minutes from the current high of 45–60 minutes by pre-verifying their documentation digitally before scheduling an appointment (Washington D.C. Department of Motor Vehicles, 2025).
- II. **Reducing On-Site Service Rejections:** Reduce the percentage of customers requested to leave the DMV because of missing documentation from 30–35% to below 15% using AI-generated mismatch summary (Washington D.C. Department of Motor Vehicles, 2025).
- III. **Improving Work Force Utilization:** Transfer the majority of document verification from 100% manual review to 20%–30% manual exception handling, thus enabling workers to complete simple tasks (i.e. restoring license plates) using the AI Co-Pilot, while focusing
- IV. their time on higher value task completion (Washington D.C. Department of Motor Vehicles, 2025).
- V. **Achieve a High Level of Automation:** The goal is to achieve a 70-80% automation rate for routine document verification through a high-efficiency, high-compliance system, where only those with a high-confidence exact match (90-100%) trigger AUTO_APPROVE.
- VI. **Increase Service Throughput:** Daily service throughput will be increased by at least 25-35% to allow the agency to serve more residents, without increasing the size of the facilities.
- VII. **Improve Public Sentiment:** There will be a 30% increase in customer satisfaction scores as a result of a more reliable, transparent and frictionless experience.

- VIII. The business goals and associated metrics align with larger-scale digital transformation efforts in public-sector organizations to improve operational efficiencies and increase civic engagement.

2.1.3. Competitive Position & Benchmark

The DC DMV is not a competitive service, however, there has been growing public expectation for digital services that have been created by the private sector. Many states have implemented online appointment scheduling, but very few have implemented an integrated agent-based artificial intelligence (AI) document confirmation system providing real-time automated feedback.

The DC DMV needs to adopt an intelligent decision support system in order to become nationally recognized as a leader in providing government services digitally. This system is being benchmarked against the best practices in public sector administrations (American Association of Motor Vehicle Administrators, 2024) and modern customer experience design (U.S. Digital Service, n.d). In addition, benchmarking against other recently completed public sector audits (North Carolina DMV) that have seen significant increases in wait times highlights the critical need for this system to process requests within a stated 15-minute goal (Enlace Latino NC, 2025).

2.1.4. Need for the System

Due to ongoing operational inefficiencies as identified through performance audits, it is essential that this system be improved immediately (District of Columbia Department of Motor Vehicles [DC DMV], 2025). The use of manual verification processes creates long customer wait times, frustration and loss of productivity both for the state and the resident (DC DMV, 2025).

By allowing customers to securely upload and verify their documents online prior to visiting, the burden of verification can be shifted upstream in the verification process. An Agentic

AI development pipeline can help significantly reduce the in-person bottlenecks in the system and reserve human staff for only edge cases (U.S. Digital Service, n.d.). This digital usage will help to reduce measurable service delays and support the DC DMV's strategic objective of providing accessible, efficient, and highly transparent services (Office of the City Administrator, 2024; Office of Management and Budget, 2022).

2.2. System Description

2.2.1. Business or Administrative Problem Addressed

The District of Columbia Department of Motor Vehicles (DC DMV) is currently experiencing many difficulties with the way they verify documents when someone applies for a driver's license or ID. The legacy system used for this process creates numerous operational and administrative headaches for both customers and agency efficiency (National Conference of State Legislatures [NCSL], 2022).

- I. Long Queue Times and Overcapacity - A customer must arrive at a DMV location in order for any verification of their documents to occur. Due to the manual method of verifying documents by staff at the counter, DMV locations have very long check-in lines and a high degree of uncertainty as to how long the wait will be to check-in and process the documents. On average, customers wait approximately 47 minutes to complete the check-in process, which is well above the acceptable service standards of most public agencies. Customers are frustrated with the long wait times and with being crowded out of the facilities, thereby leading to more delays in receiving services and adding to the operational strains placed on DMV staff.
- II. Many Incorrect / Incomplete Documents - Also, many customers arrive at the DMV with missing or invalid documents, or documents that have exceeded their expiration dates.

With no mechanism in place to pre-screen customers' documents, customers that arrive with such issues do not find this out until they reach the counter. This results in many customers being denied service, wasting staff time, re-visiting the DMV multiple times, and resulting in lower daily throughput (Pew Charitable Trusts, 2022). This inefficiency affects customer satisfaction and increases the operational workloads for DMV employees.

- III. **Bottlenecks of Document Verification are Manual Method of Performing Document Verification:** DMV is currently performing manual document verification by having employees visually reviewing documents to confirm their authenticity/validity. This process is long and time-consuming, subject to human error, and creates inconsistencies among employees performing similar tasks. Additionally, it is difficult to scale up verification during busy peak times. When evaluated internally, DMV staff spent an average of 28% of their time performing document verification tasks, thus reducing the time staff could dedicate to other, higher-value activities (Deloitte, 2021).
- IV. **Limited Utilisation of Automation and Digital Process Tools:** The DMV's current systems of record do not capitalise on contemporary technology. Therefore, DMV employees continue performing many routine matching functions that a "Digital Clerk" could perform through automated processes like Agentic AI, automated validation pipelines, or generative AI decision support. Customers are unable to verify their documents remotely and the organisation lacks real time data regarding verification trends or errors. This deficiency leaves the DMV behind other comparable governmental and commercial service organisations (Deloitte, 2021).
- V. **Ineffective Scheduling of Appointments:** Appointments can presently be made regardless of whether or not the customer has authentic documents and/or verification of such

documents. This results in both wasted appointment opportunities and no shows by customers who were not able to verify their documents before their scheduled appointment.

In addition to time spent on customers without verified documents, the absence of a digital verification gate to schedule customers optimally impedes the ability of the DMV's scheduling system to optimize resources (Pew Charitable Trusts, 2022).

- VI. **Administrative Burden & Operational Costs:** The combined result of having poor manual processes, having to make multiple trips, & having inefficient scheduling significantly increases the operational costs of the DMV. The lack of an intelligent pre-screening architecture creates additional requirements for overtime, staffing, and increased customer service demands. (NCSL, 2022; Deloitte, 2021)

2.2.2. System Operating Assumptions

The following assumptions describe the conditions we expect to be true in order to successfully implement the Agentic AI-enabled Smart DMV:

- I. **Digital Device Access by Customers:** It is assumed that DMV customers have access to a smart phone, tablet, or computer capable of capturing and uploading documents in either PDF or image format in order to conduct remote pre-verification.
- II. **Document Uploads are Readable & Authentic:** It is assumed that the customer will upload clear, legible images of their documents. The quality of the images uploaded will greatly affect the accuracy of the Google Vision OCR pipeline.
- III. **Participation of Staff in Hybrid Verification of Documents:** The DMV intends to maintain a “human-in-the-loop” model where staff review documents that are flagged as requiring exception handling by the Agentic AI using the AI Co-Pilot. This collaboration will ensure accuracy and decrease false negative results.

- IV. AI OCR Services Provide Extremely High Rates of Accuracy: The project's assumptions are that the Agentic AI engine will have an accurate-matching confidence rating of 90% or more when processing standard Government Documents which are needed for an AUTO_APPROVED Project.
- V. Reliable Systems and Internet Connections: The Project assumes both DMV staff and the customer's internet connections to be reliable and that the cloud-based infrastructure (Supabase/PostgreSQL) is expected to maintain high levels of uptime to ensure uninterrupted operations.
- VI. DMV Policy Supports Digital Pre-Verification: The assumption is that DMV leadership will approve changing policies to allow for the verification of documents digitally and asynchronously prior to providing an in-person appointment for verification.

2.2.3. Dataflow Diagrams (DFD) – Context, Level-0, Level -1

The data flow diagrams (DFDs) illustrate the movement of information within the Smart-DMV system, defining the inputs, outputs, processes, and external entities involved in the Agentic AI document verification pipeline.

- I. Context Level DFD: The Context Level DFD (*see Figure 1 in Appendix*) delineates the system boundaries and external interactions. The primary entity, the Resident, initiates the process by submitting identity and residency documents via the frontend portal. The Smart-DMV Agentic System processes these inputs and returns a "Verification Status" or "Appointment Confirmation." The DMV Admin entity receives structured "Review Packages" (including AI reasoning and mismatch summaries) and provides oversight, while the cloud Verification Database serves as the central repository for validation rules

and user records. This high-level view establishes the scope of the automated "Digital Clerk" intervention compared to the legacy manual process.

- II. Level-0 DFD: The Level-0 DFD (*see Figure 2 in Appendix*) decomposes the system into four primary subsystems: 1.0 User Authentication & Intake, 2.0 Agentic AI Extraction & Comparison, 3.0 Reasoning & Decision Logic, and 4.0 Appointment Scheduling. This diagram highlights the critical decision pathway where extracted data is verified against the Regulatory Rules Engine (D2).
 - A. High-Confidence Path (AUTO_APPROVE): Validated applications achieving an exact-match confidence score of 90–100% proceed directly to Appointment Scheduling (4.0), ensuring rapid throughput.
 - B. Exception Path (REQUIRES_STAFF_REVIEW): Submissions with mid-tier confidence scores (e.g., 60–89%) or minor fuzzy-matching discrepancies trigger a "Manual Review Request." The AI compiles a mismatched summary and routes the package to the DMV Admin, ensuring a human-in-the-loop fallback before entering the scheduling workflow.
- III. Level-1 DFD (Process 2.0 - Agentic AI Pipeline): The Level-1 DFD (*see Figure 3 in Appendix*) provides a detailed explosion of the Agentic AI Extraction & Comparison process. Incoming documents first undergo a 2.1 Image Quality Check to ensure legibility. Validated images are processed by 2.2 Google Vision OCR to convert pixel data into machine-readable JSON text. The core logic resides in 2.3 Agentic Comparison & Scoring, which routes data based on strict accuracy thresholds. Simultaneously, 2.4 PII Masking & Storage ensures that sensitive data is sanitized before being securely committed to the User Document Store (D1), adhering to strict government privacy and security requirements.

2.2.4. Business Assumptions and Risks

Risks are seen as potentially detrimental to the outcome of an AI system with regard to its function or accuracy/operation/adoption. In addition, these are considered to be elements that could provide some mitigation to the identified risks of AI systems:

- A. Misclassification and Hallucination of AI: The potential for the Agentic AI system to refuse/mark valid documents due to low quality imagery or the limitations of scanned text. If using generative AI as a co-pilot, it is also conceivable that the AI could produce 'hallucinations' about its assessment of reasoning. Solution; Human-in-the-loop fallbacks when verifying images or we will use additional textual content from our database context to strictly limit the AI's reasoning to verifiable and factually extracted pieces of information.
- B. System Downtime and Failure Rates: System interruptions due to service outages with backend cloud providers (e.g., Supabase), API call limits (e.g., Google Vision/Gemini) or internet connectivity interruptions could materially delay completing document verification processes, resulting in increased delays and operational bottlenecks for both customers and employees.
- C. Privacy Violation and Security Breach of Data: Mishandling uploaded identity documents could lead to the disclosure of sensitive Personally Identifiable Information (PII) that would legally expose the DMV for privacy violations and could result in a loss of the public trust. Solution; PII will be masked according to federal guidelines in addition to using secure, encrypted database technologies (e.g., PostgreSQL).
- D. Low Levels of User Adoption: Without widespread user use, especially amongst those who are less technologically inclined or do not have access to a reliable digital device, the

overall effectiveness of the Smart-DMV System is likely to be stymied and will force the agency to continue its current heavy reliance on face-to-face verification by many agents.

- E. **Fraudulent Use of Documents:** If users provide poor-quality or altered documents, standard automated systems for verifying documents may not flag questionable documents as fake, which would increase the risk of identity fraud. This future integration of sufficient fraud detection models and the need for continued vigilance on the part of staff for exception handling will be required to deal with this risk.
- F. **Staff Resistance and Gaps in Training:** The inability or unwillingness of staff to use the new digital Co-Pilot workflows or inadequate training delivered during the rollout of the new system will negatively impact the adoption of these new workflows, reduce productivity, and therefore not deliver the operational throughput benefits that the automation was expected to produce.
- G. **Regulatory or Compliance Changes Affecting the System:** Changes to the identity verification standards, data retention policies, or government IT policies may require an unexpected update to either system or defer the implementation date or cause additional costs to the project's on-going maintenance.

2.2.5. System Cost Targets & ROI Estimates

For implementation of the Smart-DMV project funding will be needed for both initial startup costs and ongoing operating costs. Initial costs are needed for full-stack development of the Smart-DMV system, creating a secure cloud infrastructure, utilizing APIs for Agentic AI services and OCR, and training employees. Operation Funding will be utilized to support ongoing operations such as man-hours for back-office support (e.g., establishing a business process).

Therefore, the cost estimates developed will provide a basis for evaluating the overall required level of capital to design, deploy, and maintain the Smart-DMV system.

The Smart-DMV system is also anticipated to provide measurable returns on investment through a reduction in manual work, increases in customer throughput, and improved operational efficiency of the physical facilities. Tables 3 & 4 show initial estimates of the total cost (see Table 3 in Appendix) and savings annually (see Table 4 in Appendix) associated with the implementation of the Smart-DMV Agentic solution.

3. Requirement Determination

To engineer the Smart-DMV Digital Verification System, our team utilized a focused, multi-faceted approach to capture the "As-Is" legacy state and define the "To-Be" digital architecture:

- I. Performance Audits: We analyzed operational data and identified critical bottlenecks, such as average in-person wait times of 47 minutes and a 30–35% document rejection rate at the counter.
- II. Compliance Review: We reviewed legal and technical frameworks to ensure alignment with REAL ID standards and federal mandates, specifically OMB M-23-07 (Transition to Electronic Records), prioritizing strict data security and PII masking.
- III. Stakeholder Analysis: We mapped the specific needs of two primary user groups:
 - A. Residents: Require a frictionless, upstream digital verification process to avoid physical facility turn-aways and wasted trips.
 - B. DMV Staff: Require an Agentic AI hybrid system capable of achieving a 70–80% overall automation throughput rate to reduce administrative burnout, while reserving manual review exclusively for complex exceptions via the AI Co-Pilot.

- IV. Process Mapping: We developed detailed activity diagrams (detailed in Section 1.4.1) to outline the exact, step-by-step workflows for the Customer, the Agentic AI Engine, the Staff Co-Pilot, and the System Administrators.

3.1. Technology Stack Selection

Based on the gathered requirements and the need for a highly efficient, cost-effective deployment for the capstone project, our team selected a robust, modern technology stack:

- I. Frontend & Routing: The system utilizes Vercel, providing best-in-class hosting for the React, Tailwind CSS, and Next.js web portal.
- II. Backend Database & Auth: To manage data and user sessions, the project relies on Supabase (PostgreSQL) mapped via the Prisma ORM. Supabase delivers a secure database, built-in user authentication, and S3-compatible file storage for uploaded documents in one unified package.
- III. Agentic AI & OCR Engine: The system's extraction capabilities are powered by the Google Vision API for high-performance OCR text extraction (with Paddle OCR integrated as a reliable fallback).
- IV. Reasoning & Co-Pilot: The Google Gemini API is deployed to drive the AI's core reasoning, dynamic mismatch summaries, decision-making logic, and the interactive Staff Co-Pilot features.

3.2. Requirement Gathering & Data Collection Techniques

Acquiring a real-world dataset for the Smart-DMV system presented a significant challenge. Actual identity and residency documents contain strictly protected Personally Identifiable Information (PII). Collecting or storing real resident data for testing would violate privacy rights and strict government security compliance standards.

To overcome this roadblock, the team engineered a custom Python-based synthetic data generator to create realistic, high-fidelity training data to rigorously test the system's OCR extraction and Agentic AI reasoning pipelines. To ensure the generated documents accurately mirrored the exact paperwork DC residents submit for REAL ID verification, our team utilized a standard "Residential Lease" template for Washington, DC.

This baseline template is an official, recognized form provided by the Greater Capital Area Association of REALTORS. By mapping the structural layout and terminology of this standard lease, the Python script procedurally generated a robust synthetic dataset containing thousands of variations of key data points:

- I. Standard lease variables, including the names of the "Landlord" and "Tenant," the exact address of the "Premises," and specific "Term" dates.
- II. Financial text variations, such as monthly rent installments and required "SECURITY DEPOSIT" amounts.
- III. Included DC-specific compliance addendums, such as the "Disclosure of Information on Lead-Based Paint and/or Lead-Based Paint Hazards."
- IV. Mandatory local regulatory attachments, like the "District of Columbia Tenant Bill of Rights."

This synthetic approach provided the Agentic AI engine with a high volume of accurately formatted, DC-specific documents for rigorous edge-case testing and confidence threshold calibration, completely eliminating the legal and ethical risks associated with handling real PII.

3.3. System Requirements

To efficiently and effectively move the document verification process to an earlier point in time, remove the burden from the DC DMV's day-to-day operations, and meet established

government standards for security, we have developed the Smart-DMV Digital Verification System, which must implement the following functional requirements and non-functional requirements.

3.3.1. Functional Requirements

- A. Functional requirements describe the behaviour, tasks, and actions that the system will need to perform in order to successfully facilitate the intelligent digital pre-check process.
- B. User Access and Document Upload: The system must provide a secure React-based web portal (utilizing Supabase Authentication) through which residents can access and log into their accounts, upload the required identity and residency documents (PDF or image format).
- C. Image Quality Validation: The system must perform an initial image quality check after an image has been uploaded, and will reject any images that are unreadable or severely degraded without first incurring the cost of processing them through an API backend.
- D. AI Based Text Extraction: To automatically extract text from document images uploaded by the user, the system will use the Google Vision API (with Paddle OCR as a fallback) to perform the text extraction from the uploaded images. Additionally, the result will be fed into the Agentic AI Engine for conversion to normalized JSON formatted data in order to conduct a field level comparison
- E. Confidence Scores & Logic Routing: Upon determining extracted documents against the resident's system data, the system shall evaluate and assign a confidence score, and then route accordingly based on the level of confidence:

- a. Documents with a hard, high-confidence score (90 to 100%) must result in triggering the AUTO_APPROVE status in order to provide for immediate processing.
 - b. Documents that obtain a middle confidence score and/or have required low fuzzy matches (60 to 89%) shall be assigned the status of REQUIRES STAFF REVIEW and routed into the manual workload queue.
 - c. Any document scoring below 60% shall trigger the FLAG_EXCEPTION status for either immediate rejection or deep administrative audit.
- F. Administrative Exception Handling & Staff Co-Pilot: The system shall supply a secure DMV dashboard that is used for reviewing flag downed documents by DMV staff. When opening a case, the interface shall implement a Generative-AI Staff Co-Pilot that is implemented using the Google Gemini API. Utilizing this AI solution, the Staff Co-Pilot solution shall provide for context dependent mismatch summaries, clarify the reasoning of the AI and assist staff in making manual approval or rejection determinations.
- G. Appointment Gating: The System shall strictly enforce all automated scheduling rules. Therefore, only Online Resident Users with successfully verified status (i.e., either AUTO APPROVED or Manual Approved) will be permitted to book a physical appointment slot.

3.3.2. Non-Functional Requirements

Non-functional requirements The requirements describe quality characteristics, performance objectives and stringent security criteria that the system architecture must achieve.

- I. Performance and Efficiency. The architecture for the system must decouple heavy AI processing into asynchronous (background) processes (so as to not have any timeouts at

front-end APIs), with a targeted overall automated throughput of 70–80%, in order for the project to reduce the average wait time for a customer to less than 15 minutes.

- II. **Security and Privacy.** To protect sensitive resident data, the system must enforce strict Personally Identifiable Information (PII) masking of the data to be sanitized prior to storage. Once the verification record has been finalized, it must be safely stored in an encrypted Supabase (PostgreSQL) cloud database and only accessed via Prisma ORM.
- III. **Reliability and Availability.** The architecture for the system must use highly available and serverless infrastructure (Vercel for the Next.js front-end/APIs and Supabase for back-end storage) in order to mitigate the risk of downtime and provide continuous uptime during peak usage periods.
- IV. **Regulatory and Government Compliance.** The design philosophy of the system must adhere to the U.S. Digital Service (USDS) principles and fully comply with the OMB M-23-07 mandate for the Transition to Electronic Records. Additionally, the matching logic of the system must be in full compliance with the federal REAL ID verification requirements and must continue to be in alignment with the local DC Office of the Chief Technology Officer (OCTO) data standards.

3.4. Requirement Analysis

3.4.1. Activity Diagrams (Process flow)

The UML Activity Diagram (*see Appendix, Figure 4*) visualizes the procedural logic and dynamic workflow of the Smart-DMV platform. To clearly delineate responsibilities, the process is divided into four distinct swimlanes: Resident, System AI, DMV Staff, and System Admin.

The diagram highlights the system's core value proposition through its "Happy Path" demonstrating how high-confidence OCR extractions can completely bypass the human staff lane and immediately grant the user access to the appointment calendar.

Conversely, it carefully maps the exception-handling loop, showing how low-confidence documents are flagged, routed to the DMV Staff for manual review, and either overridden or rejected back to the Resident for a re-upload.

An isolated asynchronous flow is also included in the Admin lane to demonstrate how management can dynamically adjust the AI confidence thresholds without disrupting the active user pipeline.

3.4.2. Use Cases

The core functional requirements of the Smart-DigitalVerificationDMV System from the user's perspective, as seen in the Use Case Diagram (*see Figure 5 in Appendix*), identify the following human actors – Resident, DMV Staff and System Admin. Alongside this are the external automated actors that drive the Agentic AI, namely Google Vision and Gemini APIs.

The primary goal of the Smart-DMV System is automating the verification of intelligent documents. This process occurs when a Resident submits their application, and it includes (that's included) all the aspects of the Agentic AI's data extraction, comparison and reasoning pipeline.

The Use Case Diagram also indicates the management of edge cases and the compliance with strict security requirements from government agencies by utilizing an extension (<<EXTEND>>) of the Manual Staff Review Process, which may be required to occur only under certain conditions as defined by the Agentic AI-confidentiality score is less than 90% through an exact match.

Further, the manual staff review may include (<<INCLUDE>>) consultations from the Generative AI Co -Pilot to review mismatched summary and reasoning, as well as mapping a strict business requirement that a Resident can schedule an appointment after they have been verified.

4. User Stories

4.1. Customer (Resident) User Stories:

These reflect the resident's experience using the digital-first Smart-DMV portal.

I. Document Upload & Verification:

- A. As a resident, I want to securely upload my identity and residency documents online so that I can complete verification before arriving at the DMV.
- B. As a resident, I want the system to check my document quality (clarity, legibility) immediately upon upload so that I know whether my file is acceptable before AI processing begins.
- C. As a resident, I want the Agentic AI to automatically validate my documents against the DMV database so that I don't have to wait in long lines for routine manual review.
- D. As a resident, I want to receive a proactive mismatch summary if my documents fail verification so that I understand exactly what was wrong and can correct issues before scheduling an appointment.

II. Verification Status & Notifications:

- A. As a resident, I want to receive a clear verification status (AUTO_APPROVED, REQUIRES_STAFF_REVIEW, or FLAG_EXCEPTION) so that I know exactly what to expect before my physical visit.

- B. As a resident, I want to be notified when DMV staff finalize a manual review on my flagged documents so that I can proceed with my application.
- III. Appointment Scheduling & Security:
- A. As a resident, I want the system to unlock appointment scheduling only after my documents are officially verified so that I don't waste a trip on an appointment I'm not eligible for.
 - B. As a resident, I want my sensitive Personally Identifiable Information (PII) masked and securely stored in the cloud database so that my privacy remains fully protected.

4.2. DMV Staff (Manual Reviewers) User Stories:

These reflect the human-in-the-loop workflow and interaction with the AI Co-Pilot.

- I. Manual Review & Co-Pilot Workflow:
- A. As a DMV staff member, I want to exclusively review documents that the Agentic AI flags as mid-tier confidence (60–89%) or exceptions so that I don't waste time on routine exact matches.
 - B. As a DMV staff member, I want to consult the integrated Generative AI Co-Pilot when opening a flagged case so that I can instantly read the AI's reasoning for flagging the document.
 - C. As a DMV staff member, I want to view the structured JSON mismatch summary alongside the extracted OCR text and original image so that I can quickly visually validate discrepancies.
 - D. As a DMV staff member, I want to override the AI to manually approve or reject flagged submissions so that the resident's verification status is updated accurately.

II. Exception Handling & Efficiency:

- A. As a DMV staff member, I want to add custom notes or comments to a resident's case during manual review so that future reviewers or auditors understand the decision history.
- B. As a DMV staff member, I want the automated system to handle 70–80% of routine verifications so that I can focus my time solely on complex, high-value cases requiring human judgment.

4.3. Administrator (System Manager) User Stories:

These reflect system oversight, threshold configuration, and governance.

I. System Oversight & Configuration:

- A. As an administrator, I want to dynamically configure the Agentic AI confidence thresholds (e.g., adjusting the AUTO_APPROVE exact-match threshold from 90% to 95%) so that the validation engine stays aligned with shifting DMV security policies.
- B. As an administrator, I want to update document validation rules and formatting requirements as local DC regulations change so that the logic engine remains legally compliant.
- C. As an administrator, I want to manage database user roles (Resident, Staff, Admin) via Supabase Auth so that access to PII is strictly and properly controlled.

II. Monitoring, Analytics & Compliance:

- A. As an administrator, I want to view performance dashboards showing AI automation percentages, auto-approval rates, and wait-time reductions so that I can accurately measure system ROI.

- B. As an administrator, I want to track the volume of manual reviews versus Agentic AI approvals so that I can evaluate staff workload and identify workflow bottlenecks.
- C. As an administrator, I want audit logs of all AI decisions, Co-Pilot reasoning, and manual staff overrides so that I can maintain absolute accountability and transparency for government audits.

5. Product Backlog

The product backlog consists of structured tasks executed across three primary Agile sprints. It utilizes simultaneous frontend, backend, and cloud engineering tracks to ensure the architecture matures evenly toward final production deployment.

- I. Phase 1: Project Initiation & Planning: Focuses on finalizing the project charter, defining the Agentic AI system requirements, drafting UML architectural blueprints, and provisioning the core GitHub repository and Vercel hosting environments.
- II. Phase 2 - Sprint 1: Foundational Infrastructure & UI: Establishes the core system architecture. The cloud team configures the Supabase (PostgreSQL) database, Prisma ORM schemas, and secure, PII-aware authentication. Data engineering generates the synthetic Washington DC lease datasets for AI training. Concurrently, frontend developers build the resident-facing React/Next.js upload forms and implement initial image quality pre-processing.
- III. Phase 3 - Sprint 2: Agentic AI Engine & Staff Workflows: Shifts focus to the AI core and the human-in-the-loop exception handling. The backend integrates the Google Vision API for extraction and the Gemini API for reasoning (`/lib/agentic-document-verifier.ts`). The pipeline is calibrated to send strict exact matches (90–100%) straight to auto-approval,

aiming for a total system automation throughput of 70–80%. Frontend developers build the Staff Dashboard for manual review, importantly integrating the Generative AI Staff Co-Pilot to assist with mismatch summaries and reasoning.

- IV. Phase 4 - Sprint 3: API Integration, Scheduling & Hosting: Connects all components for production. The frontend finalizes the interactive calendar, strictly gating appointment scheduling behind an "Approved" verification status. The backend configures decoupled asynchronous worker tasks to prevent API timeouts, sets up audit logs, and finalizes the deployment of the full-stack Next.js application strictly on Vercel.
- V. Phase 5: Testing & Handover: Ensures system readiness through rigorous User Acceptance Testing (UAT). This includes verifying the AI Co-Pilot's context injection to prevent hallucination, conducting security audits of PII masking and Row-Level Security (RLS) in Supabase, and load testing to guarantee < 15-minute processing times before the final Capstone handover.

6. Work-Break Down Structure

6.1. Project Phases and Tasks List

6.1.1 Phase 1: Project Initiation & Planning

Objective: Define system requirements, establish the technical environment, and map the Agentic AI architecture. Project Chartering & Requirements (see Appendix Table 5).

- Task 1 - Finalize project charter, scope, and sprint timeline.
- Task 2 - Define strict functional and non-functional requirements (including 90% AI exact-match thresholds).
- Task 3 - Draft UML Data flow, Use Case, and Activity Diagrams mapping the human-in-the-loop Agentic workflow. Technical Environment Setup

- Task 4 - Initialize GitHub repositories for the full-stack Next.js web application.
- Task 5 - Set up a Supabase project (PostgreSQL database, Auth, and Storage).
- Task 6 - Provision the serverless hosting environment exclusively on Vercel.

6.1.2 Phase 2: Sprint 1 – Foundational Infrastructure & UI

Objective: Build the resident-facing portal, establish the secure database schema via Prisma, and generate test data. Data Engineering & Synthetic Data Generation (see Appendix Table 6).

- Task 7 - Analyze the standard Washington, DC Residential Lease template.
- Task 8 - Develop a local Python script to procedurally generate synthetic DC lease documents with varied PII for AI testing.
- Task 9 - Create a labeled dataset of synthetic identity and residency documents to calibrate the OCR and reasoning engine. Database & Cloud (Supabase / Prisma)
- Task 10 - Define Prisma schema for Users, Documents, documentVerification (AI outputs), Appointments, and Audit Logs.
- Task 11 - Configure Supabase Authentication for strict role-based access (Resident, Staff, Admin).
- Task 12 - Set up Supabase Storage buckets for secure document uploads with PII masking protocols. Frontend Development (React / Tailwind / Next.js)
- Task 13 - Design and build the Landing Page highlighting the "< 15 min wait time" value proposition.
- Task 14 - Develop the secure Login and Registration components.
- Task 15 - Create the multi-step Application Form for uploading identity and residency documents. Backend Ingestion (Next.js API)

- Task 16 - Implement the initial Image Quality Check module (rejecting blurry/unreadable uploads) before AI processing.
- Task 17 - Build backend ingestion routes to securely pipe uploaded images to the OCR engine.

6.1.3 Phase 3: Sprint 2 – Agentic AI Engine & Staff Workflows

Objective: Implement the Google Vision and Gemini APIs, establish the reasoning pipeline, and build the Co-Pilot dashboard. Agentic AI Engine & Validation Logic(see *Appendix Table 7*).

- Task 18 - Integrate the Google Vision API (with Paddle OCR fallback) to extract and structure JSON text from images.
- Task 19 - Integrate the Gemini API to build the Agentic reasoning pipeline (agentic-document-verifier.ts).
- Task 20 - Implement the strict Validation Logic: route 90–100% exact matches to AUTO_APPROVE, and 60–89% to manual staff review. Admin & Staff Dashboard (Frontend)
- Task 21 - Build the Staff Dashboard UI to display the manual review queue and exception reports.
- Task 22 - Integrate the Generative AI Staff Co-Pilot interface to display the AI's structured mismatch summary and reasoning.
- Task 23 - Implement "Approve", "Reject", and "Add Comments" action buttons for manual staff overrides.
- Task 24 - Build the Admin UI to dynamically adjust AI confidence thresholds and configure rules. Data Routing & Updates (Backend / Cloud)

- Task 25 - Create endpoints to update the resident's verification status in Supabase based on AI or Staff decisions.
- Task 26 - Generate automated Exception Reports for administrative tracking.

6.1.4 Phase 4: Sprint 3 – API Integration, Scheduling & Deployment

Objective: Connect the frontend and backend, implement the appointment gateway, and deploy to Vercel production. Appointment Scheduling Integration (see Appendix Table 8).

- Task 27 - Build the interactive calendar UI component in React.
- Task 28 - Implement the "Appointment Gating" logic: restrict calendar access until verification status equals "Approved".
- Task 29 - Create backend endpoints to save and confirm appointment time slots securely. Notifications, Analytics & Background Tasks
- Task 30 - Refactor heavy Agentic AI reasoning into asynchronous background tasks to prevent frontend API timeouts.
- Task 31 - Build the Executive Dashboard displaying real-time metrics (throughput, automation rates, wait-time reductions).
- Task 32 - Implement audit logging for all AI reasoning outputs and staff/admin verification actions. Deployment & Hosting
- Task 33 - Deploy the complete Next.js full-stack application to Vercel production servers.
- Task 34 - Verify environment variables and API keys (Google, Gemini, Supabase) in the Vercel dashboard.
- Task 35 - Configure domain routing, SSL certificates, and system uptime alerts.

6.1.5 Phase 5: Testing & Handover

Objective: Ensure system stability, verify AI context constraints, and deliver the final project to the capstone sponsor. Quality Assurance & Security Testing (see Appendix Table 9).

- Task 36 - Conduct User Acceptance Testing (UAT) for the resident upload flow and the Co-Pilot staff review flow.
- Task 37 - Perform security audits to ensure PII masking and Supabase row-level security (RLS) policies are functioning.
- Task 38 - Test AI context-injection boundaries to verify the Co-Pilot does not hallucinate reasoning outside of the database records. Documentation & Handover
- Task 39 - Compile the final capstone project report (incorporating updated Agentic architecture, WBS, and UML diagrams).
- Task 40 - Create user manuals for the DMV Staff Dashboard and System Administrators.
- Task 41 - Final presentation and handover of the system repository to Sponsor Dr. Soheil Sibdari.

6.2. Major Deliverables

- I. Web Portal (React / Tailwind / Next.js): The responsive, resident-facing frontend for secure authentication, digital document uploads, and gated appointment scheduling.
- II. Cloud Database & Storage (Supabase / Prisma): The secure PostgreSQL database and file storage system managing user accounts, verification statuses, and PII-masked documents, accessed strictly via the Prisma ORM.

- III. **Agentic AI Pipeline (Vision & Gemini APIs):** The intelligent backend engine that utilizes Google Vision for high-fidelity OCR text extraction and the Gemini API for advanced reasoning, structuring the extracted text into normalized JSON data.
- IV. **Validation Logic Module:** The automated decision engine (`agentic-document-verifier.ts`) that enforces strict government security thresholds, routing exact matches (90–100%) straight to auto-approval, while generating structured mismatch summaries for edge cases.
- V. **Staff Co-Pilot & Admin Dashboard:** The secure interface for DMV staff to manually process flagged exceptions. This deliverable includes the integrated Generative AI Co-Pilot to explain the AI's reasoning, alongside tools for administrators to monitor overall system throughput and adjust confidence thresholds.

7. Schedule Structure

7.1. Task Dependencies

The Smart-DMV Digital Verification System utilizes a concurrent engineering approach, relying on strict Finish-to-Start (FS) dependencies across the frontend, backend, and cloud tracks to ensure seamless integration. The project's critical path is dictated by four major dependency milestones:

- I. **Infrastructure & Data Prerequisites (Sprint 1):** The foundation of the system relies on the cloud track. The Supabase database schema and role-based authentication (Tasks 10–11) must be fully provisioned before the frontend can securely accept document uploads. Simultaneously, the procedural generation of synthetic DC lease datasets (Tasks 7–9) acts as a strict prerequisite before the AI team can begin training and calibrating the Google Vision extraction and Gemini reasoning engines.

- II. Agentic AI & Workflow Integration (Sprint 2): The frontend Staff Dashboard development (Tasks 21–23) is highly dependent on the backend AI logic. The Agentic AI pipeline (agentic-document-verifier.ts) and strict 90% confidence-scoring algorithms (Tasks 18–20) must be operational first. The Staff UI and integrated Co-Pilot rely entirely on retrieving the structured JSON mismatch summaries generated by the AI for low-confidence exceptions.
- III. Status-Gated Scheduling (Sprint 3): The final resident user experience relies on a strict API dependency. The interactive appointment calendar (Tasks 27–29) cannot be implemented or accessed until the backend validation endpoints successfully push an "Approved" verification status to the Supabase database.
- IV. Testing & Deployment (Phase 5): Final system validation is dependent on full-stack integration. Comprehensive User Acceptance Testing (UAT) and load testing (Tasks 36–38) including verifying the AI Co-Pilot's context boundaries to prevent hallucination are strictly dependent on the successful deployment of the complete Next.js full-stack application to Vercel production servers (Tasks 33–35).

This dependency structure ensures that data flows logically from initial user upload, through automated Agentic AI processing, and into final human-in-the-loop staff review without creating development bottlenecks.

7.2. Project Task Schedule

7.2.1. Network Diagram

The Project Evaluation and Review Technique (PERT) Network Diagram (*see Appendix, Figure 6*) illustrates the chronological flow and logical dependencies of the Smart-DMV Digital Verification System. Rather than a simple linear progression, the

project architecture is designed around three parallel engineering tracks—Frontend, Backend (Agentic AI), and Cloud (Database)—that advance simultaneously through the Agile sprints and merge at critical integration milestones.

Key highlights of the project's network flow include:

- I. Parallel Execution (Sprint 1): Foundational development kicks off concurrently. The resident-facing UI (Next.js/React) is built alongside the Supabase Database and Authentication setup (Cloud). The Prisma database schema must be established before the data engineering team can generate the synthetic lease datasets used for testing.
- II. Strategic Convergence (Sprints 2 & 3): The independent tracks merge as the core Agentic reasoning logic is introduced. The Google Vision extraction and Gemini API reasoning pipelines (Backend) connect with the verification status endpoints (Cloud), which then dictate the exception-handling queue and Co-Pilot interface displayed on the Staff Dashboard (Frontend). All three tracks fully converge at the strict API Appointment Gating milestone before final production deployment to Vercel.

7.2.2. Critical Path

The critical path for the Smart-DMV Digital Verification System represents the longest sequence of dependent tasks that must be completed on schedule to ensure the project finishes by the final capstone deadline. Because this sequence has zero slack (or float), any delay in these specific tasks will directly delay the final system handover.

While the frontend User Interface (UI) and administrative dashboards are vital to the project, they are largely asynchronous and possess schedule flexibility. The true critical

path is anchored by the backend Agentic AI and Cloud data pipeline, flowing through the following sequence:

Project Business Case → Prisma DB Schema & Auth Setup → Generate Synthetic Data → Agentic AI Pipeline (Vision & Gemini) → API Appointment Gating → Vercel Deployment → Final UAT & Handover.

- I. Phase 1 & Sprint 1 (Architecture & Infrastructure): The project cannot begin technical execution until the Supabase database schema (via Prisma) and secure, role-based authentication are fully provisioned. Furthermore, the AI models cannot be tested or refined until the data engineering team successfully generates the synthetic identity and DC lease datasets.
- II. Sprint 2 (Agentic AI Pipeline Integration): This is the most complex technical milestone. The Google Vision OCR extraction and the Gemini-powered reasoning logic (agentic-document-verifier.ts) must be perfectly functional before any downstream verification rules or Co-Pilot features can be applied.
- III. Sprint 3 (API Appointment Gating): The frontend scheduling calendar is completely blocked until the backend validation endpoints successfully push "Approved" statuses to the Supabase database.
- IV. Phase 5 (Deployment & UAT): The project culminates in the final Next.js full-stack deployment to Vercel production servers, which must be live before User Acceptance Testing (UAT) including verifying AI context boundaries and the final sponsor handover can occur.

By identifying this path, our team knows to prioritize backend Agentic AI infrastructure and cloud endpoints above all other tasks, ensuring the core intelligent verification engine is never waiting on frontend cosmetic updates.

7.2.3. Timeline and Milestones with Gantt Chart

The project timeline and key deliverables (*see Appendix, Figure 7,8,9*) are visually represented in the Gantt chart, providing a comprehensive schedule of the Smart-DMV development lifecycle. The chart maps all 41 project tasks across the five project phases, illustrating the concurrent execution of the frontend, backend, and cloud engineering tracks over the course of the semester.

This visual timeline allows our team to actively monitor progress, manage overlapping dependencies, and ensure strict adherence to academic and developmental deadlines.

8. Detailed Software/Systems Design

8.1. Classes and methods

8.1.1. Core Classes from Problem Domain

- I. Class Name: User : I am including the User class because secure access is the foundation of the entire system. Every workflow begins with authentication, and the requirements clearly describe role based access for resident, staff, and administrator users. A shared User class allows us to centralize identity and permissions before any document upload, review, or configuration activity takes place.

- A. Methods: `authenticate()`, `authorizeRole()`, `updateAccountStatus()`

- B. These methods are necessary because we must validate credentials, enforce role permissions, and manage account state to protect sensitive identity data and system integrity.
- II. Class Name: Resident/Customer : I am including the Resident/Customer class because the resident is the primary actor in the system. The resident initiates applications, uploads required documents, receives verification outcomes, and requests appointment scheduling only after approval. This reflects the core workflow described in our system introduction and user stories.
 - A. Methods: startApplication(), uploadDocument(), viewVerificationStatus(), requestAppointmentScheduling()
 - B. These methods align directly with the resident journey. The resident must create a case, submit documents into the intake pipeline, monitor status, and move toward scheduling once verification is complete.
- III. Class Name: StaffReviewer : I am including the StaffReviewer class because the system uses a human in the loop model. When OCR confidence is low, cases are routed to staff for manual review. This exception handling pathway is explicitly required in our functional scope.
 - A. Methods: viewManualReviewQueue(), reviewSubmission(), approveSubmission(), rejectSubmission(), addCaseNotes()
 - B. These methods reflect the responsibilities of staff. They must retrieve flagged cases, inspect extracted data alongside the original document, make approval or rejection decisions, and document their reasoning for traceability.

- IV. Class Name: Administrator : I am including the Administrator class because our system includes governance and configuration responsibilities. The requirements state that administrators manage roles, configure verification settings, and monitor dashboards and audit logs. This is a distinct responsibility from staff review.
- A. Methods: manageUserRoles(), configureVerificationSettings(),
 viewDashboards(), reviewAuditLogs()
 - B. These methods support oversight and policy management. They allow administrators to control access, adjust thresholds, and monitor system performance and compliance.
- V. Class Name: Application : I am including the Application class because our workflow revolves around a resident case. The application connects the resident, the requested service, the required documents, the verification state, and eligibility for scheduling. Modeling this as a single lifecycle object keeps the system structured and traceable.
- A. Methods: create(), submitForVerification(), updateStatus(),
 isEligibleForScheduling()
 - B. These methods represent the lifecycle of a case. We create it, push it into verification, update its state based on decisions, and determine whether it can move forward to scheduling.
- VI. Class Name: Document : I am including the Document class because document intake and processing are the core system functions. Residents upload identity and residency documents, and the system validates image quality, runs OCR, applies privacy controls, and stores the result securely.

- A. Methods: `upload()`, `validateImageQuality()`, `preprocessForOCR()`, `maskPII()`, `linkToApplication()`
 - B. These methods reflect the document lifecycle from ingestion through preparation for automated validation. They ensure that uploaded content is usable, compliant, and properly associated with the correct application.
- VII. Class Name: `OCRRun` : I am including the `OCRRun` class because OCR is not just a background function. It is a defined processing event that produces extracted data and a confidence score, and our routing logic depends on that score.
 - A. Methods: `executeExtraction()`, `calculateConfidence()`, `setRunStatus()`, `produceReviewPackage()`
 - B. These methods allow us to run extraction, compute confidence, track processing completion, and prepare results for either automated decisioning or manual review.
- VIII. Class Name: `VerificationDecision` : I am including the `VerificationDecision` class because every application must reach a clear and accountable outcome. The system supports both automated approval and manual approval or rejection with notes. A decision object allows us to represent that outcome formally and traceably.
 - A. Methods: `autoDecide()`, `recordManualDecision()`, `attachReason()`, `commitOutcome()`
 - B. These methods allow decisions to be created consistently, whether automated or manual, and ensure that reasons are captured before updating the application state.

- IX. Class Name: Appointment : I am including the Appointment class because scheduling is explicitly gated by verification success. The system must prevent unverified residents from booking slots, and it must manage reservation and confirmation states.
- A. Methods: checkEligibility(), reserveSlot(), confirm(), cancel()
 - B. These methods enforce the scheduling gate, manage time slot allocation, and support appointment lifecycle changes.
- X. Class Name: AuditLog : I am including the AuditLog class because accountability and compliance are critical in an identity verification context. Administrators must be able to review verification actions and system activity.
- A. Methods: logEvent(), queryByCase(), queryByUser(), exportForCompliance()
 - B. These methods ensure that system actions are recorded and retrievable for oversight and reporting.
- XI. Class Name: NotificationService : I am including the NotificationService class because residents must receive feedback about verification outcomes and appointment confirmations. Clear communication reduces repeat visits and improves operational efficiency.
- A. Methods: notifyStatusChange(), notifyCorrectionNeeded(), notifyAppointmentConfirmation(), notifySystemAlert()
 - B. These methods support real time feedback and keep both residents and administrators informed about important system events.

8.1.2. Supporting Classes from Problem Domain

- I. Class Name: ServiceType : I am including ServiceType as a supporting class because different DMV services may require different documents and rules. Modeling service types allows us to scale and adjust requirements without changing the core structure.
 - A. Methods: getRequiredDocumentTypes(), getApplicableRules()
 - B. These methods allow the selected service to determine what documents and validation logic apply.
- II. Class Name: ServiceCenter : I am including ServiceCenter as a supporting class because appointments occur at specific locations. Scheduling must consider location capacity and service availability.
 - A. Methods: listAvailableSlots(), validateCenterServiceSupport()
 - B. These methods allow us to expose available slots and ensure that a service is supported at the chosen location.
- III. Class Name: ApplicationStatus : I am including ApplicationStatus as a supporting class to standardize the vocabulary of states such as approved, needs correction, and manual review. This ensures consistent logic across the system.
 - A. Methods: isSchedulingEligibleStatus(), isTerminalStatus()
 - B. These methods help determine whether a case can move forward or has reached completion.
- IV. Class Name: ApplicationStatusHistory : I am including ApplicationStatusHistory to support traceability over time. Tracking state changes strengthens accountability and helps staff understand case progression.

- A. Methods: recordChange(), getTimeline()
 - B. These methods allow us to capture and retrieve a full status history for review and compliance purposes.
- V. Class Name: DocumentType : I am including DocumentType to distinguish between identity and residency docuToRequiredFields(), validateFormatExpectations()
 - A. These methods support structured OCR and validation tailored to the document category.
- VI. Class Name: DocumentVersion : I am including DocumentVersion to support resubmissions without losing history. Residents may upload corrected documents, and we need to track current and previous versions.
 - A. Methods: createNewVersion(), setAsCurrent(), listVersions()
 - B. These methods allow us to preserve document history while clearly identifying the active version for decisioning.
- VII. Class Name: VerificationRule : I am including VerificationRule to allow administrators to configure thresholds and validation logic rather than hard coding rules. This increases flexibility and governance control.
 - A. Methods: evaluate(), updateThreshold(), activateDeactivate()
 - B. These methods support rule execution, modification, and lifecycle control.
- VIII. Class Name: RunRuleResult : I am including RunRuleResult to capture which rules passed or failed during evaluation. This strengthens explainability and improves manual review efficiency.
 - A. Methods: recordResult(), summarizeFailures()

- B. These methods allow us to store structured outcomes and provide clear feedback when verification fails.
- IX. Class Name: OCRField : I am including OCRField to define which extracted data points are important to the system. This supports structured extraction and consistent validation.
 - A. Methods: defineField(), markAsKeyField()
 - B. These methods standardize what data we extract and prioritize during verification.
- X. Class Name: OCRFieldValue : I am including OCRFieldValue to store normalized extracted values and per field confidence scores. This supports more precise routing and clearer manual review.
 - A. Methods: storeExtractedValue(), normalize(), assignFieldConfidence()
 - B. These methods allow us to maintain structured, scored extraction outputs that improve decision quality, transparency and to allow validation logic to vary by type.

8.2. Class Diagram

This class diagram (*see Figure 10 in the Appendix*) outlines the robust architecture for a comprehensive AI-powered document verification and scheduling platform, specifically tailored for a system like a modernized DMV. At its foundation is a role-based access control structure, branching from a central User class into specialized roles: Resident/Customer, Staff/Reviewer, and Administrator. These users interact with the core workflows driven by the Application and Appointment classes.

A significant portion of the design models an automated processing pipeline as it handles file management via Document and DocumentVersion, performs structured text extraction represented by OCRResult and OCRField, and evaluates the parsed data against configured VerificationRule parameters to generate a VerificationDecision. The system also ensures strict compliance and user communication through tightly integrated AuditLog, ApplicationStatusHistory, and NotificationService components.

8.3. Reports

The reporting capability is essential for system oversight and performance measurement, as illustrated (*see Appendix, Figure 11*). These reports are automatically generated for administrative tracking, providing the DMV management with essential insights into the system's operational efficiency, including metrics on verification rates, automation percentages, and error trends.

Specifically, the Exception Reports allow administrators to monitor documents flagged as low-confidence (20–30% score) that required manual review, which is critical for evaluating staff workload and assessing the project's return on investment (ROI).

8.4. Business Rules (for Screens/Fields)

The Smart DMV Digital Verification System is governed by essential business rules (*see Appendix, Table 10*) enforced across all interfaces to maintain security, compliance, and process efficiency. Access to the system is protected by Security Rules that lock an account after five failed login attempts, and Complexity requirements mandating a minimum of eight characters and one number for passwords.

The document upload process is tightly controlled by Format rules that only permit PDF, JPG, or PNG files under 10MB, and an AI Check to ensure acceptable image quality, preventing unreadable documents from entering the verification pipeline.

Crucially, the system enforces a strict Precondition for service delivery: appointment scheduling is entirely disabled until a resident's verification status is officially Approved. On the administrative side, Role-Based Access ensures Staff Reviewers only see assigned or flagged cases, and Admin Configuration rules dictate that the AI Confidence Threshold must remain within the range of 0.50 and 0.99, with all changes to this threshold being logged via an Audit Rule.

8.5. Cost-Benefit Analysis

The operations evaluation (*see Table 11 in Appendix*), including a comparison between a Smart DMV digital verification system and a traditional DMV verification model, is the underlying purpose of this evaluation. The use of Agentic AI-based digital workflows versus traditional reactive, manual, in-person verification processes provides operational efficiencies, cost reductions and increased service delivery.

8.5.1. Operational and Workflow Comparison

In the traditional DMV model, verification of documents happens at the time of the visit, creating large numbers of unprepared visits, inefficient use of employee time and extreme congestion in DMV facilities. With the Smart DMV digital verification system, document uploads and verification occur prior to scheduling an appointment, resulting in a change to a proactive, digital-first workflow for document verification.

This new upstream verification model creates significant performance improvements in key operational metrics.

- The throughput of completed transactions increases from approximately 400 to 475 per 500 customers, representing an increase in throughput of 18.75%. The DMV can therefore serve a higher volume of customers without increasing physical capacity due to this throughput increase. Additionally, it reduces operational idle time by optimally utilizing available appointment capacity to achieve this increase in throughput and produce corresponding financial savings.
- Reducing the number of unprepared or failed visits: The number of unprepared or failed visits are reduced from about 20% to about 5%, which represents a 75% reduction overall. This directly eliminates wasted appointment time and reduces the number of repeat visits.
- Cost Efficiency: The cost of a successful visit is reduced by about \$1.67 to about \$0.85 per visit, which represents a 51% reduction in administration costs on a per-transaction basis.
- Automation Impact: By leveraging the 90% confidence level of the Agentic AI Pipeline, the Smart DMV System achieved an estimated automation rate of approximately 70% to 80% for documentary verifications in Year 1 and 75% to 90% in Year 2, allowing employees to devote more of their time to handling complex exceptions through Generative AI Co-Pilot.

Overall, the Smart DMV System takes the current verification process and makes it a predictable and scalable workflow that eliminates many of the currently existing operational bottlenecks while enhancing the reliability of distributed service.

8.5.2. Financial Impact

Finally, The overall financial benefit of the Smart DMV System will be achieved through improved levels of operational efficiency; specifically through reduced levels of failure-based appointments and through enhanced transaction volumes.

- The average savings per user (~\$0.82) are calculated from the reduction of failed appointments and the overall reduction of expenditures associated with failed visits and their associated administrative timeframes. When failing visits drop from 20% to 5%, this translates into approximately 15 percentage points newly employed as a result of previously wasted appointments. Each appointment that was precluded was previously cash in the employee's pocket in forms of time savings, resource utilization, and increased efficiency associated with scheduling.
- Aggregate Savings (assuming 500 users/day):
 - Daily Savings: Approximately \$409 per day.
 - Annual Savings (Gross): Approximately \$149,000.
- Cost Structure:
 - Year 1 (Implementation Phase):
 - Development and infrastructure: \$28,000 (one-time).
 - Operations (hosting, maintenance, API usage): \$22,000.
 - Total Cost: \$50,000.
 - Year 2+ (Operational Phase):
 - Ongoing operations: \$22,000 annually.
- Net Benefit & Return on Investment (ROI):
 - Year 1: Net Benefit of ~\$99,000, resulting in a 198% ROI.

- Year 2+: Net Benefit of ~\$127,000, resulting in a sustained ~577% ROI.
- Break-even Analysis: The system achieves break-even within approximately 4 months of deployment, indicating rapid recovery of initial investment and early realization of net positive returns or financial benefits.

8.5.3. Key Strategic Takeaways

The Smart DMV system is a strong short-term financial return and has a longer-term strategic value beyond direct dollar savings.

- I. **High Operational Efficiency Impacts:** In the first year of implementation, the system will automate 70-80% of document verification tasks, with Year Two being 75-90%. This answers repetitively performed manual work and redistributes worker effort towards more complex and valuable tasks.
- II. **Scalable Service Delivery:** All throughput increases will be achieved without the need for staffing or physical expansion. Therefore, the DMV can successfully manage growing volumes of product/service requests.
- III. **Sustained Long-Term Financial Returns:** After the initial deployment stage, the system will operate on a low-cost, high return on investment basis, in its fourth year, will generate an average annual net benefit of more than \$120,000.
- IV. **Improved Customer Experience:** The system's pre-visit verification process eliminates uncertainty for residents, enables fewer repeat visits, and creates a more predictable and streamlined service experience.
- V. **Alignment with Digital Government Programs:** The Smart DMV will position the agency as a leader in the use of AI-enabled, consumer-focused service delivery

models, as well as providing alignment with larger governmental modernization and digital transformation initiatives.

8.6. Strategic Plan to Implement Agentic AI

8.6.1. Agentic AI for Document Verification Workflow

- A. The Smart DMV system has an Agentic AI Document Verification Workflow for automated verification of documents submitted by residents of the State of Maryland.
- B. This document verification workflow will use Agentic AI to validate documents submitted by residents of Maryland in a smart, efficient manner rather than through traditional rule based validation systems.
- C. Using an intelligent agent allows for reasoning, decision-making, and explanation generation to assist with the validation of documents submitted by residents of Maryland.
- D. The integration of Agentic AI into the document verification process will improve efficiency and transparency.
- E. The Smart DMV system is able to validate documents accurately and reliably by combining Optical Character Recognition (OCR) processing, Structured Data Comparison (SDC), Agentic AI Reasoning, and Human in the Loop workflows.

8.6.2. System Methodology and Workflow Process

The methodology for this system is a rigid Pipeline of Decisions that has a Multi-Step structure. The stages of the Data Process are: Extract & Submit to Application, Compare to Other Applications, Reason To Determine Approval/Rejection, Make The Approve/Rejection Decision, and Provide An Explanation.

Data Processing Steps:

- A. Residents Upload Document(s) (PDF or Image) With Their Application Data;
- B. The System Uses OCR to Extract Text From The Uploaded Document, Formats Those Field Format Into JSON, Processes The Data, and The Result(s) Are Stored In A Database.
- C. Staff Pulls Pending Verifications Case Using Appropriate API.
- D. Staff Reviews The Case And Updates Are Posted Back To The Database (Approval, Rejection, Or Notes).

8.6.3. Agentic Behavior

- A. The function of the AI engine is to standardise and clean extracted data.
- B. The AI engine creates a series of field-by-field comparisons where it looks for differences between the system data that was provided and the OCR document data.
- C. Using these comparisons, the AI engine assigns a confidence score and generates an explanation.
- D. Finally, the agent provides the ultimate result of either producing no rejection (AUTO APPROVE), needing staff review (REQUIRES STAFF REVIEW), or flagging it as an exception (FLAG EXCEPTION).

8.7. Implementation Architecture & Algorithms

8.7.1. Applied Algorithms, Models, and Equations

- A. The Google Vision API is used by the system to obtain text from the uploaded document of a resident.

- B. The components of a resident's information (name, address, city, zip code) are compared to the tenant name, lease dates, address, and landlord name found in the document with extracted fields.
- C. The system uses algorithms based on case-insensitive string matching of resident documents as well as fuzzy logic for general validation of fields such as names and addresses.
- D. However, zip codes must match exactly.
- E. The Google Gemini API is used to generate contextual responses through AI for Co-Party Staff in real-time.

8.7.2. App/Web Framework Implementation Details

- A. Backend Logic: Custom verification logic is housed within `/lib/agentik-document-verifier.ts`.
- B. API Layer: The architecture includes four primary APIs:
 - a. Document Submission API (POST `/api/verification/submit`) to accept documents and trigger the decision workflow.
 - b. Staff Queue API (GET `/api/staff/verifications`) to retrieve cases, apply filters, and auto-escalate timed-out cases.
 - c. Staff Decision API (POST `/api/staff/verifications/review`) to handle approve/reject decisions and store staff notes.
 - d. AI Assistant API (POST `/api/staff/verifications/assistant`) to handle staff queries using the Gemini API.
- C. Database Integration: The backend uses Supabase (PostgreSQL) mapped via the Prisma ORM.

- D. Schema Design: The core table is documentVerification, which captures fields such as status, aiConfidence, aiExplanation, mismatchSummary, ocrText, and reviewNotes.

8.8. Design Decisions & Project Differentiators

8.8.1. Decision Transparency & Human-in-the-Loop

- A. Transparency: The system creates an organized summary of all mismatched fields, their values, whether they matched or not, and the reason(s) for the mismatch, in a JSON structure. With this information, staff members will have a better understanding of why they are being automated and gain confidence in the system.
- B. Balanced automation: Staff will be responsible for dealing with edge cases, while the AI will deal with the majority of cases routinely.
- C. Staff co-pilot: As soon as a case is opened, a real-time AI Co-Pilot appears, providing context-aware assistance, decision support, reasons behind which fields did/do not match, and risk reasoning using case data and Optical Character Recognition (OCR) results.

8.8.2. Scalability and Future-Proofing

- A. Architecture: The API-driven architecture allows for large-scale processing of documents while supporting easily expandable AI behavior.
- B. Future-readiness: The architecture design provides for the addition of multi-document verification (cross-validation), fraud detection models, advanced Natural Language Processing (NLP) reasoning, and the addition of a feedback loop for learning from staff decisions.

9. Sprint 1

9.1. Sprint Backlog

Sprint 1 lays the crucial technical groundwork for the Smart-DMV Digital Verification System. The primary objective of this sprint is to build the core infrastructure required for secure user onboarding, document ingestion, and early data preparation prior to the Agentic AI integration. To achieve this, our team executed tasks across four concurrent tracks: cloud architecture, data engineering, backend ingestion, and frontend development.

Cloud Architecture & Database Setup Our cloud team established the system's secure foundation using Supabase. This involved configuring role-based Supabase Authentication to manage Resident, Staff, and Admin access securely. Simultaneously, our team translated the system's Entity-Relationship Diagram (ERD) (*see Appendix, Figure 12*) into a live PostgreSQL database schema utilizing the Prisma ORM.

This schema explicitly defines the relational structures for User, ResidentProfile, Application, Document, and the foundational tables for future Appointment and DocumentVerification records. Finally, secure storage buckets were configured with strict Row-Level Security (RLS) policies to ensure that sensitive Personally Identifiable Information (PII) within uploaded documents remains protected and strictly isolated.

Data Engineering & Synthetic Datasets To support the intelligent components arriving in Sprint 2, our data engineering team initiated the creation of vital training assets. By analyzing the standard Washington, DC Residential Lease template, the team developed local Python scripts to procedurally generate synthetic identity and residency documents.

This synthetic labeled dataset is a vital prerequisite, allowing the team to safely train, calibrate, and benchmark the Agentic AI pipeline (Google Vision extraction and Gemini

reasoning) without exposing real citizen data during development. (Dataset Google Drive Link: https://drive.google.com/file/d/1ujScRAxaYA8BDaWswgRBgiJhGE8inbNq/view?usp=share_link)

Backend: The Image Quality Gatekeeper Before documents are routed to the complex Agentic AI engine, the Next.js backend team built an initial document ingestion API route that acts as a strict "Quality Check Gatekeeper" (see *Appendix, Figure 13*). This lightweight preprocessing layer evaluates incoming uploads for minimum clarity standards. It applies essential image-processing validations to ensure the file is ready for OCR.

If an image is severely blurred, heavily shadowed, or corrupted, the gatekeeper immediately rejects the upload and prompts the resident to try again. This critical step ensures that only high-quality, readable documents are passed to the expensive AI APIs in Sprint 2, maximizing exact-match accuracy and reducing unnecessary computational load and API token costs. (GitHub Repository: <https://github.com/Srinivas-Vengaldas/Smart-DMV>)

Frontend Development & Deployment On the frontend, the team delivered the first iteration of the resident-facing web interface, successfully deploying the Next.js/React application to the project's Vercel production environment. This serverless deployment enables early usability testing and provides a tangible interaction layer for the system APIs (viewable at: <https://v0-smart-dmv.vercel.app/>).

The frontend deliverables include three core views:

- **The Landing Page:** This interface introduces residents to the Smart-DMV portal and provides clear navigation to core services, including document verification, driver licensing, and vehicle registration. It establishes the system's modern user-experience

foundation and clearly communicates the value proposition of reducing DMV wait times (*see Appendix, Figure 14*).

- **Secure Registration Interface:** This component integrates directly with Supabase Authentication, reflecting the sprint's focus on secure user onboarding. The form captures essential account credentials and establishes the resident's profile within the database, preparing the system to intelligently pre-fill downstream application workflows (*see Appendix, Figure 15*).
- **Multi-Step Application Workflow:** This structured form allows residents to securely enter the specific personal details and biometric data (e.g., height, weight, eye color) required by the official DC DMV REAL ID application. By breaking the application into progressive steps, the UI minimizes cognitive load, reduces input errors, and cleanly transitions the user into the crucial document-upload stage (*see Appendix, Figure 16*).

9.2. Sprint Burndown Chart

The Sprint Burndown Chart (*see Appendix, Figure 17 and Table 12*), tracks daily progress against the planned workload, providing a visual comparison of completed tasks versus the ideal completion trajectory. By monitoring the actual burndown line against this baseline, our team continuously assesses development velocity and identifies emerging bottlenecks such as unexpected delays in database schema provisioning or frontend deployment.

This agile artifact facilitates real-time scope management, ensuring that foundational infrastructure, data engineering tasks, and initial user interface components are delivered strictly on schedule to support the critical-path AI integration in the subsequent sprint.

10. Sprint 2

10.1. Sprint 2 Backlog

The completion of Sprint 2, on Agentic AI Engine Integration and staff workflows, leads us to a fully functioning AI component of the Smart-DMV Digital Verification System. The Google Vision API has been successfully integrated into the AI engine for high-definition text extraction, and the Gemini API powers the core reasoning pipeline of our AI engine.

The Validation Logic Module was additionally calibrated to ensure strict compliance with government security requirements by automatically approving only exact matches (90%-100% confidence) to achieve our goal of achieving 70%-80% of our overall automation through this validation process.

Exceptions now route through the staff dashboard, which has been completely built out and now uses the Generative AI Staff Co-Pilot to help staff members review manual processes through structured JSON data sets.

Additionally, the team capitalized on high development velocity by developing the API integration and appointment scheduling logic ahead of schedule. As such, sprint 3 key deliverables were successfully started ahead of schedule.

10.2. Sprint 2 Burndown Chart

The Sprint Burndown Chart (*see Appendix, Table 13 and Figure 18*), tracks the progress of the development team's work during the Sprint period by visually comparing the Ideal Burndown (the planned trajectory for work completion) against the Actual Burndown (the remaining effort day-by-day).

The chart facilitates continuous scope management by ensuring that foundational infrastructure, data engineering tasks, and initial user interface components are delivered strictly on schedule to support the critical-path AI integration in the subsequent sprint.

11. Sprint 3

11.1. Sprint 3 Backlog

The Sprint 3 backlog contained essential items for Phase 4 Release, which needed to connect the frontend and backend architectures to further develop in terms of a full production deployment. Each task was prioritized based on how much it affected the final system functionality, User Experience and User Experience, as well as preparing for handover of the Capstone.

The high-priority items were aimed at closing the user journey loop (i.e. building the interactive React calendar), along with enabling the appointment verification gating logic requirements to be strictly enforced (i.e. employing a strict appointment verification gating logic).

Supporting and medium-priority tasks were focused on achieving critical enterprise features, such as converting the Agentic AI reasoning from heavy synchronous processes into asynchronous background processes (to prevent timeouts when calling APIs from the frontend), configuring automated email/SMS notifications, building out the Executive Analytics Dashboard, and deploying the final version of the full-stack application to the Vercel production servers.

11.2. Sprint 3 Burndown Chart

For this deployment-focused sprint, the team planned and successfully executed 26 total story points over a 21-day period (Mar 24 – Apr 13). Because Sprint 3 centered on connecting previously built architectures, the actual burndown (*see Table 14 and Figure 19*) experienced a highly efficient, steady drop.

The backend scheduling endpoints and frontend gating logic (Tasks 27, 28, and 29) were completed rapidly in the first week. The most significant plateau occurred during the execution of the 5-point Executive Dashboard (Task 31). However, the sprint concluded with a strong finish as the final asynchronous background tasks (Task 30) and Vercel environment configurations (Tasks 33, 34, and 35) were deployed securely, successfully bringing the Agentic AI platform to a production-ready state for final testing.

12. Final Product Showcase & Deployment

12.1 Narrated Product Demo (Video Link)

A comprehensive video demonstration of the Smart-DMV Digital Verification System is available below. The video walks through the end-to-end lifecycle of a document, showcasing the resident upload experience, the real-time Agentic AI extraction and scoring (via Google Vision), and a demonstration of the Generative AI Staff Co-Pilot assisting a manual review.

- Demo Video Link :

https://drive.google.com/file/d/1BhLORmGoSsdUuJM2pX2C3_uz37yVHZcc/view?usp=sharing

12.2. GitHub Source Code Repository

The complete source code for the full-stack Next.js application, including the Prisma database schemas and the Agentic AI logic pipeline (agentic-document-verifier.ts), is hosted securely on GitHub.

- Repository Link: <https://github.com/EnochSakaEugene/smart-dmv>

12.3. Application Access (Deployment Link & Login Instructions)

The production environment is hosted on Vercel, with the PostgreSQL database securely managed by Supabase.

- Production URL : <https://v0-smart-dmv.vercel.app/>
- Test Credentials :
 - Resident Persona: Username: test_user@example.com , Password:
DemoPass123!
 - DMV Staff Persona: Username: staff1@smartdmv.gov ,Password: Staff@123
 - System Admin Persona: Username: admin1@smartdmv.gov ,Password:
Admin@123

13. Product Testing

13.1. Functionality Testing Steps

Functionality testing focused on the backend API endpoints and the strict rules engine governing the Agentic AI:

- Extraction Accuracy: Tested the Google Vision API against the procedurally generated synthetic dataset of Washington, DC residential leases to ensure high-fidelity text extraction across various lighting and rotation conditions.
- Logic Thresholds: Verified that the 90% exact-match threshold correctly triggered the AUTO_APPROVE state without false positives, strictly enforcing government security standards.
- Co-Pilot Context Boundaries: Conducted edge-case testing on the Gemini API prompt injection to ensure the Co-Pilot generated mismatched summaries only based on the database and OCR text, successfully preventing AI hallucinations.

- PII Security: Audited Supabase Row-Level Security (RLS) policies to confirm that uploaded documents and masked PII were completely inaccessible to unauthorized user roles.

13.2. Usability Testing

Usability testing (User Acceptance Testing - UAT) evaluated the human-computer interaction for both primary user groups:

- Resident UI: Testers navigated the multi-step Next.js application form. Feedback resulted in adding clearer loading spinners during the AI processing phase to reduce user abandonment.
- Staff UI: DMV staff personas tested the manual review dashboard. The side-by-side layout (original image next to the Co-Pilot mismatch summary) was refined to minimize scrolling and eye strain, successfully proving that the tool reduces manual review time per document from minutes to seconds

14. Marketing Plan

As the Smart-DMV platform approaches Minimum Viable Product (MVP) status, the marketing strategy pivots from development-focused messaging to GovTech positioning. The system is marketed as a transformative enterprise solution centered on trust, regulatory compliance, and technological differentiation.

14.1. Marketing Objective & Positioning

- Primary Objective: Establish Smart-DMV as a leading reasoning-based automated verification system for regional government services by the end of Q2.
- Core Value Proposition: A paradigm shift from static, rule-based validation to Agentic AI-driven reasoning.

- **Positioning Statement:** The platform delivers a dual capability—automating routine cases while providing explainable decision-support for complex edge cases, ensuring strict traceability and accountability required in high-compliance government environments.

14.2. Target Audience

- **Economic Buyers:** DMV/IT Directors, Chief Technology Officers, and municipal Procurement Officers. They prioritize budget allocation, systemic ROI, and long-term strategic adoption.
- **Operational Influencers:** DMV Staff Managers and Administrative Leaders. They evaluate system usability, staff efficiency gains, and day-to-day operational impact.

14.3. Integrated Marketing Strategy

The overarching campaign theme is “Reason, Not Rules.” This highlights the system’s key differentiator: providing intelligent, explainable decision-making. By emphasizing transparency and human-in-the-loop collaboration, the strategy directly addresses and mitigates common concerns surrounding "black box" AI adoption in the public sector.

14.4. Marketing Channels & Execution

- **High-Touch Sales & Demos:** Focus on live demonstrations of the Staff Co-Pilot handling edge cases (e.g., complex address mismatches) where traditional OCR fails.
 - **KPIs:** Sales-qualified leads, demo completion rates, contract conversion timelines.
- **Content Marketing:** Establish thought leadership via a white paper titled “From Static Data to Agentic Intelligence: Why Rules-Based DMV Automation is Obsolete,” accompanied by a technical blog series detailing fuzzy matching and reasoning logic.
 - **KPIs:** Gated content downloads, blog traffic, lead generation rates.

- Industry Events: Secure speaking engagements at GovTech conferences pitching “AI with a Justification Layer.” Emphasize explainable outputs and mismatch summaries.
 - KPIs: Booth engagement, consultation bookings.
- Targeted Social Media (LinkedIn): Deploy B2B campaigns targeting GovTech professionals using the core message: “90% automated, 100% justified.”
 - KPIs: Click-through rates, open rates, engagement levels.

15. Project Retrospection & Future Recommendations

15.1. Retrospection (What went well / What went wrong)

- I. What Went Well: The decision to pivot to a modern Next.js and Supabase stack allowed for rapid, secure iteration. Furthermore, substituting basic OCR with the Agentic AI pipeline (Google Vision + Gemini) drastically elevated the project from a simple web form to an enterprise-grade automation tool. The synthetic data generation proved invaluable for safe AI training.
- II. What Went Wrong / Challenges: Tuning the confidence thresholds proved challenging. Initially, the AI was too strict, rejecting minor typos and pushing too much volume to the manual staff queue. Fine-tuning the fuzzy-matching logic for addresses while maintaining strict security on ID numbers required significant backend refactoring. Additionally, managing asynchronous API timeouts on Vercel required shifting heavy AI workloads to background tasks.

15.2. Recommendations for the Future ("Notes to self")

- I. Monitor API Costs: Integrating dual-layered AI (Vision for extraction, Gemini for reasoning) incurs token costs. Future iterations must closely monitor API usage spikes and implement caching where appropriate.

- II. Maintain the Synthetic Dataset: Do not rely on real user data for future testing. The Python-based synthetic lease generator is a massive asset; keep it updated with new document templates as local DC regulations change.
- III. Continuous CI/CD: Maintain strict GitHub branch protections and automated testing pipelines to ensure that future frontend updates do not accidentally break the core AI routing logic.

15.3. Future Enhancements (Beyond Current MVP)

- I. Multilingual Co-Pilot: Expand the Generative AI Co-Pilot and resident portal to seamlessly support Spanish, Amharic, and other prominent languages in the DC metro area to improve accessibility.
- II. Federal Database Integration: Move beyond OCR verification by securely hooking the backend directly into federal datasets (such as the SAVE program for immigration status) to automate REAL ID compliance fully.
- III. Native Mobile Application: Package the Next.js frontend into a React Native mobile application, allowing residents to use their smartphone cameras natively for higher-resolution document scanning and direct biometric (FaceID) authentication.

References

American Association of Motor Vehicle Administrators. (2024). Virginia added to AAMVA's Digital Trust Service.

<https://www.aamva.org/publications-news/aamva-news/virginia-added-to-aamva%E2%80%99s-digital-trust-service>

District of Columbia Department of Motor Vehicles. (n.d.). DC DMV document

Verification guide.<https://dmv.dc.gov/page/dc-dmv-document-verification-guide>

District of Columbia Department of Motor Vehicles. (2025). Agency performance oversight hearing fiscal years 2024-25. Government of the District of Columbia.

<https://dmv.dc.gov/release/dc-department-motor-vehicles-agency-performance-oversight-hearing-fiscal-years-2024-25>

Deloitte. (2021). Digital government transformation: Modernizing public services for improved citizen experience. Deloitte Insights.

[https://www2.deloitte.com/us/en/insights/industry/public-sector/digital-transformation-in-government.html\(www2.deloitte.com in Bing\)](https://www2.deloitte.com/us/en/insights/industry/public-sector/digital-transformation-in-government.html(www2.deloitte.com in Bing))

Enlace Latino NC. (2025, January). Audits reveal long waits and low morale at North Carolina DMV offices.

<https://enlcelatinonc.org/en/Audits-reveal-long-waits-and-low-morale-at-North-Carolina-DMV-offices./>

Executive Office of the President.

https://www.whitehouse.gov/wp-content/uploads/2022/12/m_23_07-m-memo-electronic-records_final.pdf

National Conference of State Legislatures. (2022). Improving DMV customer service through modernization. NCSL. [https://www.ncsl.org/transportation/dmv-customer-service-](https://www.ncsl.org/transportation/dmv-customer-service-modernization)

[modernization](https://www.ncsl.org/transportation/dmv-customer-service-modernization) (ncsl.org in Bing)

Office of Management and Budget. (2022). Update to transition to electronic records (M-23-07).

National Conference of State Legislatures. (n.d.). DMV modernization and digital service delivery. <https://www.ncsl.org/transportation/dmv-modernization>

Office of the City Administrator. (2024). FY 2026 performance plans. Government of the District of Columbia.

<https://oca.dc.gov/sites/default/files/dc/sites/oca/FY26%20Plan%20-%20OCA.pdf>

Office of the District of Columbia Auditor. (2022). DMV customer service operations audit. <https://dcauditor.org/reports>

Office of the City Administrator. (2024). Agency performance oversight hearing fiscal years 2024-25. Government of the District of Columbia.

<https://oca.dc.gov/sites/default/files/dc/sites/oca/FY26%20Plan%20-%20OCA.pdf>

Pew Charitable Trusts. (2022). How states can improve digital service delivery. Pew Research Center. <https://www.pewtrusts.org/en/research-and-analysis/reports/2022/05/how-states-can-improve-digital-service-delivery> (pewtrusts.org in Bing)

StateScoop. (2024). NASCIO releases top 10 state CIO priorities for 2025.

<https://statescoop.com/state-cio-technology-priorities-2025-nascio-list/>

U.S. Digital Service. (n.d.). Digital services playbook.

<https://playbook.usds.gov/>

U.S. Digital Service. (n.d.). Improving government services through digital transformation. <https://www.usds.gov/what-we-do>

Appendix

Table 1

Project Charter

Project Title	DC DMV Digital Document Verification System
Project Scope	The scope of this project is to develop and implement a web-based application that enables residents of Washington D.C. to upload and pre-validate their identity and residency documents using a secure Agentic AI document verification process. The application will work as an electronic "Digital Clerk" and use Google Vision OCR and an intelligent reasoning engine to check that all documents meet Washington D.C. DMV standards prior to allowing residents to make an appointment. By moving the verification process "upstream," it will significantly reduce the time spent processing verifications in person.
Product Description / Deliverables	The final product will consist of the following integrated modules: A. Web Portal: A responsive frontend built with React and Tailwind CSS for seamless document upload and immediate user feedback. B. Agentic AI Engine ("Digital Clerk"): A backend service that utilizes OCR to extract text from images and pipes it into a multi-step decision pipeline (Extract → Compare → Reason → Decide → Explain). C. Staff Co-Pilot & Admin Dashboard: A secure interface allowing DMV staff to review complex cases, featuring a real-time Generative AI assistant (Co-Pilot) that provides structured mismatch summaries and explains the AI's reasoning. D. Verification Database: A secure cloud storage environment for validated metadata, featuring mandatory Personally Identifiable Information (PII) masking.
Project Objectives	The project's success will be measured against the following targeted metrics: A. Reduce Wait Times: Cut average in-person wait times from 45–60 minutes down to < 15 minutes. B. Increase Automation: Achieve a 70–80% automated verification rate via the Agentic AI reasoning engine. C. Decrease Rejections: Lower the on-site document rejection rate from the current 35% down to < 15%. D. Optimize Staffing: Shift the manual verification workload from 100% manual review to only 20–30%, utilizing a human-in-the-loop strategy for exception handling.

Business Case	The manual verification of documents at the DC DMV creates significant delays and inefficiencies in terms of daily operations. Approximately 30-35% of all transactions are denied for documentation errors resulting in wait times over 45 minutes on average per in-person transaction and contributing to facility and community congestion. The objectives outlined in this proposal are in compliance with OMB Memorandum M-23-07 (the transition to electronic records) and the USDA's mandate to improve the way governments provide services to the public. A complete electronic and Agentic AI-based digital verification model will provide the DMV with a mechanism for measuring improvements to their operational throughput, staffing efficiencies, and trust by citizens.	
Pre-assigned Resources	A. Faculty Advisor: Dr. Soheil Sibdari B. Development Tools: React, Tailwind CSS, TypeScript, Supabase , Prisma API, Supabase Storage, Google Vision, Vercel, Github. C. Budget: Capstone Student Tier (Free).	
Stakeholder List		
Name	Title	Role/Responsibility
Dr. Soheil Sibdari	Project Sponsor	Provides academic oversight, approves charter, and evaluates at every step of production.
Srinivas Vengaldas	Project Manager	Responsible for OCR Algorithm Engine and Backend modules. Manages schedule, scope, development, and risk mitigation.
DC Residents (Simulated)	End Users	Primary users of the document upload portal; source of UAT feedback.
DMV Administration	Business Owner	Beneficiaries of the operational improvements (simulated entity).
Enoch Eugene Saka	Scrum Master	Responsible for coding frontend/backend modules.

Summary Milestone Schedule

Description	Due Date
<u>Project Preparation</u>	
Finalize Charter, Set up Dev Environment	Feb 8, 2026
<u>Collecting Requirements</u>	
Finalize DFDs, User Stories, Tech Stack Selection	Feb 15, 2026
<u>Design Phase</u>	
UI/UX Wireframes, Database Schema, AI Logic Design	March 1, 2026
Frontend Build, API Integration, Backend Logic (Implementation / Coding)	March 20, 2026
UAT, Security Checks, PII Masking Verification (Testing & Validation)	April 15, 2026
<u>Final Presentation</u>	
(Project Handover, Capstone Defense)	April 27, 2026
<i>Note. Please refer to page no.7.</i>	

Table 2*Key Performance Metrics for Proposed DMV System*

Metric	Current System (As-Is)	Proposed System (To-Be)	Expected Impact
Average customer wait time	45–60 minutes	< 15 minutes	Faster service, reduced congestion
Agentic AI Auto-Approval Rate	0%	70-80%	Drastically reduces routine workload and increases daily service capacity.
Customers turned away due to documentation	30–35%	< 15%	Fewer repeat visits
Manual document verification workload	100% manual	20–30% manual	Staff efficiency improvement
Document verification automation	0–10%	70–80%	Faster processing
Daily service throughput	Baseline	+25–35%	More customers served
Customer satisfaction	Low / inconsistent	+30% improvement	Higher public trust

Note. Please refer to page no. 10

Table 3*Initial System Cost Targets*

Cost Category	Description	Estimated Cost
System Development & Integration	Front-end and back-end development, database design, AI integration, API configuration, and testing.	\$80,000–\$120,000 (one-time)
Cloud Hosting & Infrastructure	Compute resources, storage for uploaded documents, load balancing, monitoring, and scalability.	\$20,000–\$100,000 per year
AI OCR & Document Analysis Services	OCR processing, confidence scoring, fraud-detection checks.	\$10,000–\$15,000 per year
Security & Compliance	Encryption, access controls, audit logging, and compliance reviews.	\$10,000–\$15,000 (initial)
Training & Change Management	Staff onboarding, training materials, and transition support.	\$8,000–\$12,000 (initial)
Ongoing Maintenance & Support	Bug fixes, updates, performance tuning, and user support.	\$15,000–\$25,000 per year

Note. Please refer to page no. 19

Table 4*Initial Return On Investment Estimates*

Benefit Category	Description	Estimated Annual Savings
Reduced Staff Overtime	Automation of up to 80% of document verification reduces overtime hours.	\$60,000–\$80,000
Increased Throughput & Efficiency	Faster processing and fewer bottlenecks increase daily service capacity.	\$50,000–\$70,000
Lower Turn-Away & Repeat Visit Costs	Reducing incorrect documentation by 50% decreases wasted appointments.	\$30,000–\$40,000
Improved Customer Satisfaction	Fewer complaints and escalations reduce administrative overhead.	Qualitative benefit
Total Estimated Annual Savings	Combined operational savings from automation and efficiency gains.	\$150,000–\$200,000

Note. Please refer to page no. 19

Table 5*Phase 1: Project Initiation & Planning*

ID	Task	Priority	Points
1	Finalize project charter and timeline.	High	2
2	Define strict functional and non-functional requirements (including 90% AI exact-match thresholds).	High	3
3	Draft UML Data flow, Use Case, and Activity Diagrams mapping the Agentic workflow.	High	3
4	Initialize GitHub repositories for the full-stack Next.js web application.	High	1
5	Set up a Supabase project (PostgreSQL database, Auth, Storage).	High	2
6	Provision the serverless hosting environment exclusively on Vercel.	High	2

Note. Please refer to page no. 31

Table 6*Phase 2: Sprint 1 – Foundational Infrastructure & UI*

ID	Task	Priority	Points
7	Analyze the standard Washington, DC Residential Lease template.	Medium	1
8	Develop a local Python script to procedurally generate synthetic DC lease documents for AI testing.	High	5
9	Create a labeled dataset of synthetic identity and residency documents.	High	2
10	Define Prisma schema for Users, Documents, documentVerification (AI outputs), and Appointments.	High	3
11	Configure Supabase Authentication for strict role-based access.	High	3
12	Set up Supabase Storage buckets for secure document uploads with PII masking protocols.	High	3
13	Design and build the Landing Page highlighting the "< 15 min wait time".	Low	2
14	Develop the secure Login and Registration components.	High	3
15	Create the multi-step Application Form for uploading documents.	High	5
16	Implement the Image Quality Check module.	High	3
17	Build Python pre-processing scripts.	High	5

Note. Please refer to page no. 31

Table 7*Phase 3: Sprint 2 – AI Engine Integration & Staff Workflows*

ID	Task	Priority	Points
18	Integrate the Google Vision API (with Paddle OCR fallback) to extract and structure JSON text from images.	High	5
19	Integrate the Gemini API to build the Agentic reasoning pipeline (agentic-document-verifier.ts).	High	5
20	Implement the strict Validation Logic: route 90–100% exact matches to AUTO_APPROVE.	High	5
21	Build the Staff Dashboard UI to display the manual review queue and exception reports.	High	3
22	Integrate the Generative AI Staff Co-Pilot interface to display the AI's structured mismatch summary and reasoning.	High	3
23	Implement "Approve", "Reject", and "Add Comments" action buttons for manual staff overrides.	High	2
24	Build the Admin UI to dynamically adjust AI confidence thresholds and configure rules.	Medium	3
25	Create endpoints to update the resident's verification status in Supabase based on AI or Staff decisions.	High	3
26	Generate automated Exception Reports for administrative tracking.	Medium	3

Note. Please refer to page no. 32

Table 8*Phase 4: Sprint 3 – API Integration, Scheduling & Hosting*

ID	Task	Priority	Points
27	Build the interactive calendar UI component in React.	High	3
28	Implement the "Appointment Gating" logic: restrict calendar access until verification status equals "Approved".	High	3
29	Create backend endpoints to save and confirm appointment time slots securely.	High	3
30	Refactor heavy Agentic AI reasoning into asynchronous background tasks to prevent frontend API timeouts.	Medium	3
31	Build the Executive Dashboard displaying real-time metrics (throughput, automation rates, wait-time reductions).	Medium	5
32	Implement audit logging for all AI reasoning outputs and staff/admin verification actions.	Medium	3
33	Deploy the complete Next.js full-stack application to Vercel production servers.	High	2
34	Verify environment variables and API keys (Google, Gemini, Supabase) in the Vercel dashboard.	High	2
35	Configure domain routing, SSL certificates, and system uptime alerts.	Medium	2

Note. Please refer to page no. 33

Table 9*Phase 5: Testing & Handover (April 2026)*

ID	Task	Priority	Points
36	Conduct User Acceptance Testing (UAT) for the resident upload flow and the Co-Pilot staff review flow.	High	5
37	Perform security audits to ensure PII masking and Supabase row-level security (RLS) policies are functioning.	High	3
38	Test AI context-injection boundaries to verify the Co-Pilot does not hallucinate reasoning outside of the database records.	Medium	3
39	Compile the final capstone project report (incorporating updated Agentic architecture, WBS, and UML diagrams).	High	5
40	Create user manuals for the DMV Staff Dashboard and System Administrators.	Medium	3
41	Final presentation and handover of the system repository to the Sponsor.	High	5

Note. Please refer to page no. 34

Table 10*Smart-DMV Digital Verification System Business Rules*

Field	Rule Type	Rule Description	Error/Action
Login / Registration Screen			
Authentication	Required	Users must be authenticated via Supabase Auth before accessing protected screens.	Redirect to Login
Failed Attempts	Security Rule	Lock account after 5 failed login attempts.	Account Locked Message
Email	Format/Unique	Must be valid format and unique at registration.	Show 'Invalid or already used email'
Password	Complexity	Minimum 8 characters with at least 1 number.	Show 'Password requirements not met'
Service Selection Screen			
Service Selection	Required	Users must select exactly one service type.	Show 'Select one service type'
Checklist Mapping	System Rule	The checklist must match the selected service type.	Load matching checklist
ServiceType	Allowed Values	Must be New, Renewal, or Registration.	Show 'Invalid service type'

Document Upload Screen (Customer)

Upload	Required	Document upload required before verification begins.	Disable Submit
File Type	Format	Only PDF/JPG/PNG allowed.	Show 'Invalid file format'
File Size	Range	Maximum 10MB.	Show 'File too large'
Image Quality	Pre-processing	Next.js Gatekeeper rejects severely blurred/dark images before routing to AI APIs.	Status = Needs Reupload
DocumentType	Required	Must select Proof of ID, Residency, etc.	Show 'Document type required'
E-Sign Agreement	Required	Users must check 'I Agree' before submitting.	Prevent Submission

Verification Status / Feedback Screen

Status	Allowed Values	Must be AUTO_APPROVE, REQUIRES_STAFF_REVIEW, or FLAG_EXCEPTION.	System enforced
AI Feedback	Required if Flagged	Must display the Generative AI structured mismatch summary explaining the error.	Enable Re-upload Option
Manual Review	Notification	Display 'Staff reviewing via Co-Pilot' and next step info.	Send Notification

Appointment Scheduling Screen (Customer)

Verification Check	Precondition	Scheduling strictly disabled until status = AUTO_APPROVE or manually approved by staff.	Disable Calendar / Button
Time Slots	Availability	Only available slots displayed (no double booking).	Hide booked slots
Active Appointment	Limit Rule	Only one active appointment per application.	Prevent Duplicate Booking
Location	Required	Must select from predefined DMV locations.	Show 'Select location'

Staff Review Dashboard

Case Visibility	Role-Based Access	Staff only see documents scoring < 90% exact-match confidence.	Access Restricted
Co-Pilot Review	System Rule	Must display original document alongside Vision API JSON extraction and Gemini Co-Pilot reasoning.	Required Review Step
Audit Log	System Rule	Record who approved/rejected, the AI's reasoning, and the timestamp in Supabase.	Log Entry Created
Decision	Required	Staff must select Override Approve or Confirm Reject.	Cannot Submit
Comments	Conditional	Staff must provide a written justification if overriding the AI's recommendation.	Show 'Enter comment'

Admin Configuration Screen

Access Control	Role-Based	Only Admin roles can access via Supabase RLS.	Access Denied
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Threshold Changes	Audit Rule	Any dynamic change to the Agentic AI thresholds must be logged.	Audit Trail Created
Confidence Threshold	Range	Must strictly enforce 90–100% for AUTO_APPROVE and 60–89% for REQUIRES_STAFF_REVIEW.	Show 'Invalid threshold value'
Reports Screen (Admin/Management)			
Report Access	Role-Based	Staff see operational queue; Admin sees full throughput analytics.	Access Restricted
Date Range	Required	Date range must be selected to fetch background task data.	Show 'Select date range'
Export Format	Allowed Values	Export limited to PDF or CSV.	Show 'Invalid export format'

Note. Please refer to page no. 48

Table 11*Cost-Benefit Summary Table*

ITEM	YEAR 1	YEAR 2+	NOTES
Development & Infrastructure	\$28,000	—	One-time system build, AI integration, deployment.
Operations (Hosting, Maintenance)	\$22,000	\$ 22,000	Cloud hosting, API usage, support. Annual recurring.
Total Cost	\$50,000	\$ 22,000	
Automation Rate	70-80%	75-90%	Agentic AI-driven document verification.
Savings per User	~ \$0.82	~ \$0.82	Derived from reduced failed visits and processing time.
Daily Savings (500 users/day)	~ \$409/day	~ \$409/day	Based on throughput improvements and failure reduction.
Annual Savings (Gross)	~ \$149,000	~ \$149,000	Total operational savings before costs.
Net Benefit	~ \$99,000	~ \$127,000	Annual savings minus total costs.
ROI	198%	~ 577% (sustained)	$(\text{Net Benefit} \div \text{Cost}) \times 100$.
Cost per Successful Visit	\$1.67 => \$0.85	\$0.85	51% reduction maintained post-implementation.
Throughput (per 500 users)	~ 400 => ~ 475	~ 475	+18.75% increase in completed transactions.
Failure Rate	~ 20% => ~ 5%	~ 5%	75% reduction in unprepared visits.

Breakeven Point ~ 4 months — Time to recover initial investment.

Note. Please refer to page no. 49

Table 12*Sprint 1 Burndown Points*

Date	Day	Ideal Burndown	Actual Burndown	Completed Tasks
Feb 9	Day 0	35	35	Sprint Kickoff
Feb 10	Day 1	32.5	34	Task 7 (1 pt)
Feb 11	Day 2	30	32	Task 9 (2 pts)
Feb 12	Day 3	27.5	29	Task 10 (3 pts)
Feb 13	Day 4	25	24	Task 8 (5 pts)
Feb 14	Day 5	22.5	21	Task 11 (3 pts)
Feb 15	Day 6	20	21	(Weekend)
Feb 16	Day 7	17.5	21	(Weekend)
Feb 17	Day 8	15	18	Task 12 (3 pts)
Feb 18	Day 9	12.5	16	Task 13 (2 pts)
Feb 19	Day 10	10	13	Task 14 (3 pts)
Feb 20	Day 11	7.5	8	Task 15 (5 pts)
Feb 21	Day 12	5	8	(Weekend)
Feb 22	Day 13	2.5	5	Task 16 (3 pts)
Feb 23	Day 14	0	0	Task 17 (5 pts)

Note. Please

refer to page no. 59

Table 13*Sprint 2 Burndown Points*

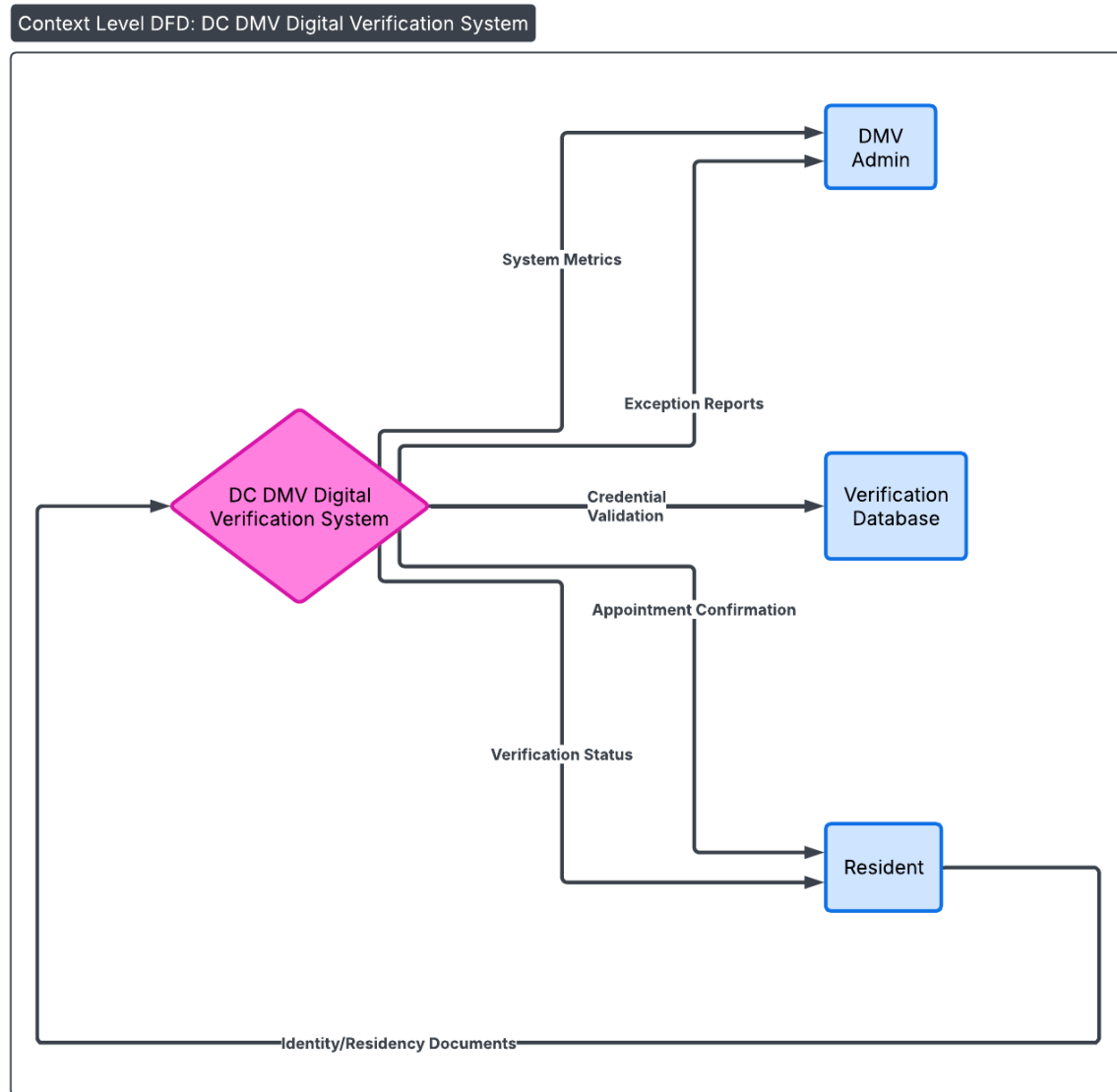
Date	Day	Ideal Burndown	Actual Burndown	Completed Tasks
Feb 24	Day 0	32.0	32	Sprint Kickoff
Feb 25	Day 1	29.7	32	
Feb 26	Day 2	27.4	29	Task 21 (3 pts)
Feb 27	Day 3	25.1	27	Task 23 (2 pts)
Feb 28	Day 4	22.8	24	Task 22 (3 pts)
Mar 1	Day 5	20.5	24	(Weekend)
Mar 2	Day 6	18.2	24	(Weekend)
Mar 3	Day 7	16.0	19	Task 18 (5 pts)
Mar 4	Day 8	13.7	16	Task 25 (3 pts)
Mar 5	Day 9	11.4	13	Task 24 (3 pts)
Mar 6	Day 10	9.1	10	Task 26 (3 pts)
Mar 7	Day 11	6.8	10	(Weekend)
Mar 8	Day 12	4.5	10	(Weekend)
Mar 9	Day 13	2.2	5	Task 19 (5 pts)
Mar 10	Day 14	0.0	0	Task 20 (5 pts)

Note. Please refer to page no. 60

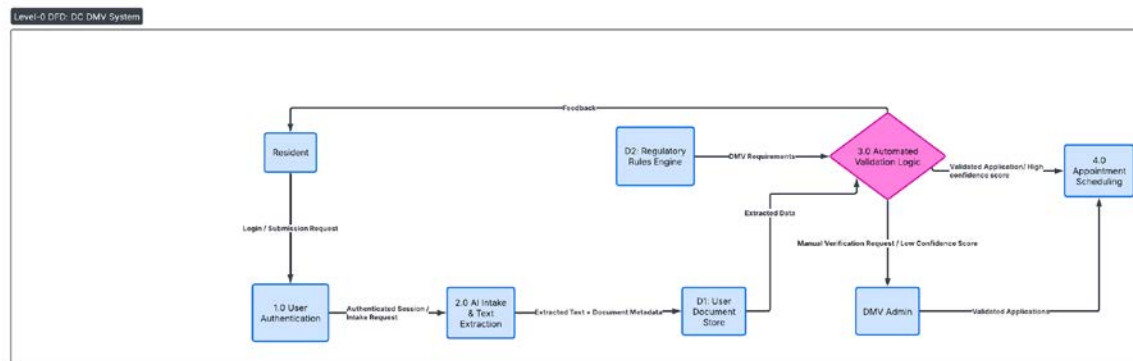
Table 14*Sprint 3 Burndown Points*

Date	Day	Ideal Burndown	Actual Burndown	Completed Tasks
Mar 24	Day 0	26	26	Sprint Kickoff
Mar 26	Day 2	23.4	23	Task 27 (3 pts)
Mar 29	Day 5	19.5	20	Task 28 (3 pts)
Apr 1	Day 8	15.6	17	Task 29 (3 pts)
Apr 4	Day 11	11.7	14	Task 30 (3 pts)
Apr 6	Day 13	9.1	12	Task 33 (2 pts)
Apr 8	Day 15	6.5	10	Task 34 (2 pts)
Apr 10	Day 17	3.9	5	Task 31 (5 pts)
Apr 12	Day 19	1.3	2	Task 32 (3 pts)
Apr 13	Day 20	0	0	Task 35 (2 pts)

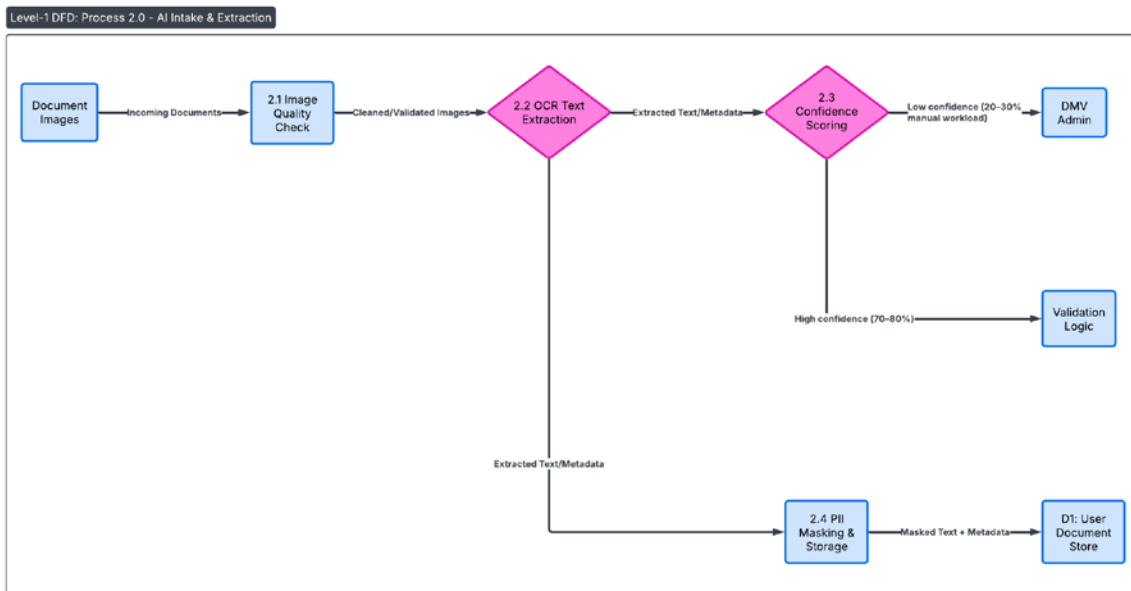
Note. Please refer to page no. 61

Figure 1*Context Level Dataflow Diagram*

Note. Please refer to page no. 16.

Figure 2*Level 0 Dataflow Diagram*

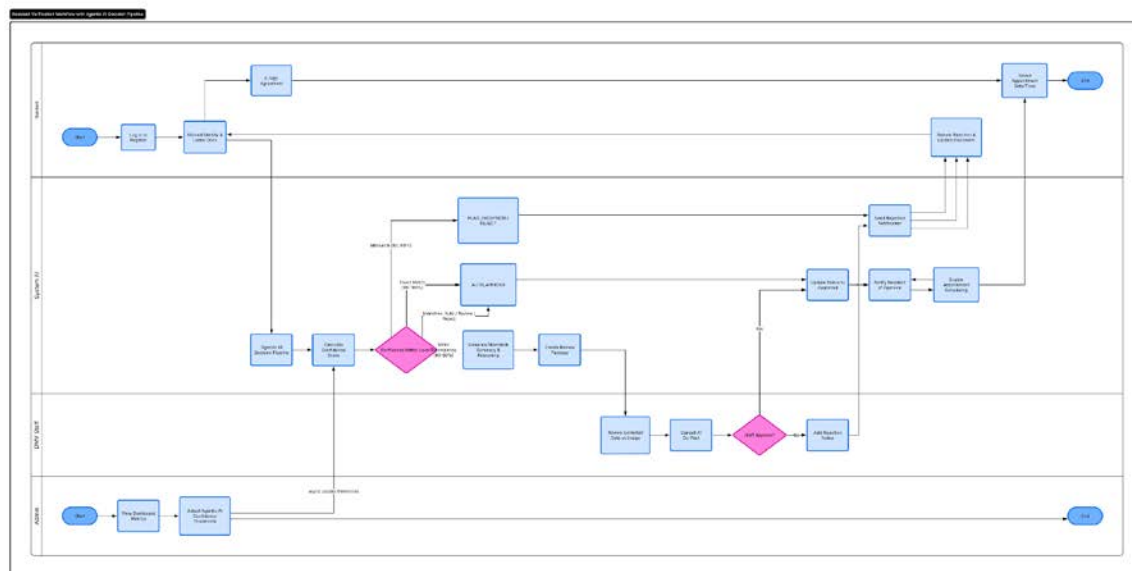
Note. Please refer to page no. 16

Figure 3*Level 1 Dataflow Diagram*

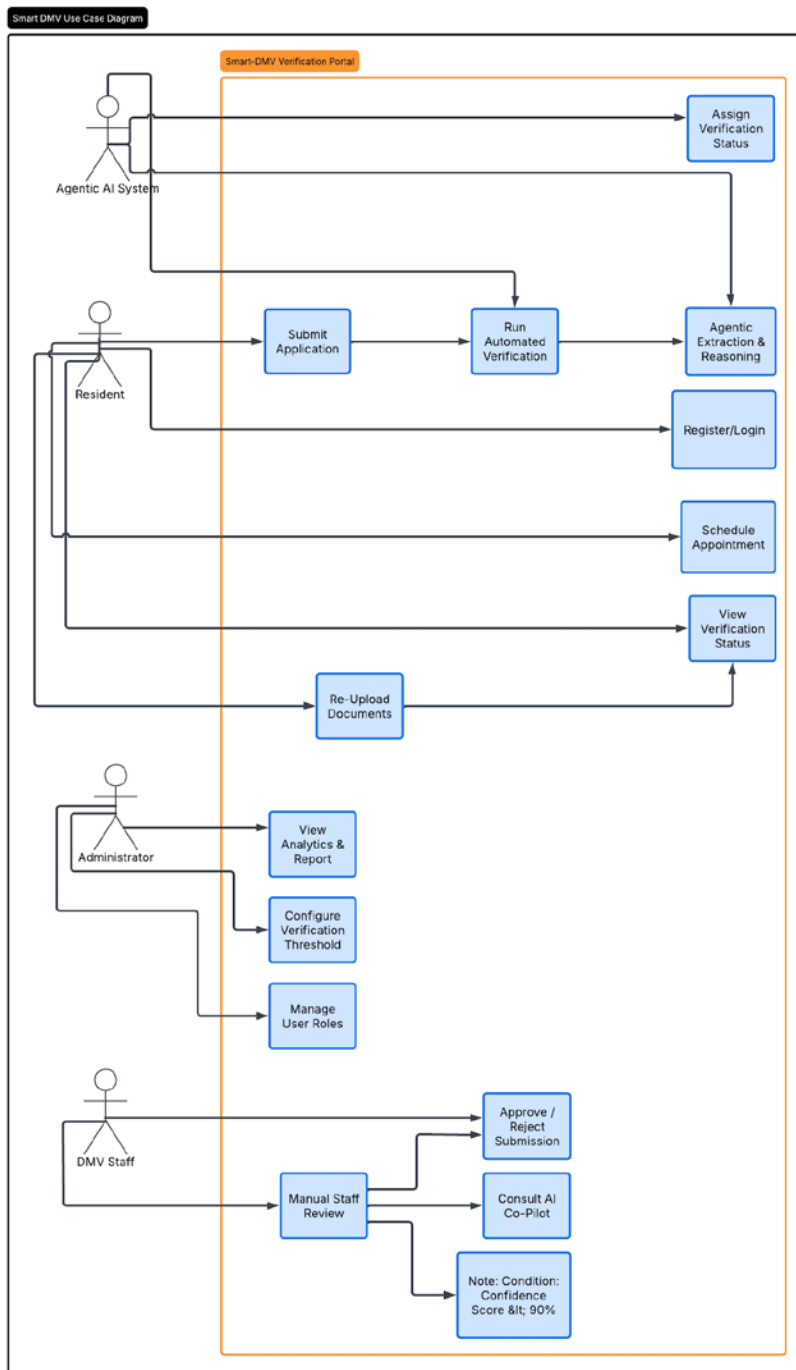
Note. Please refer to page no. 17

Figure 4

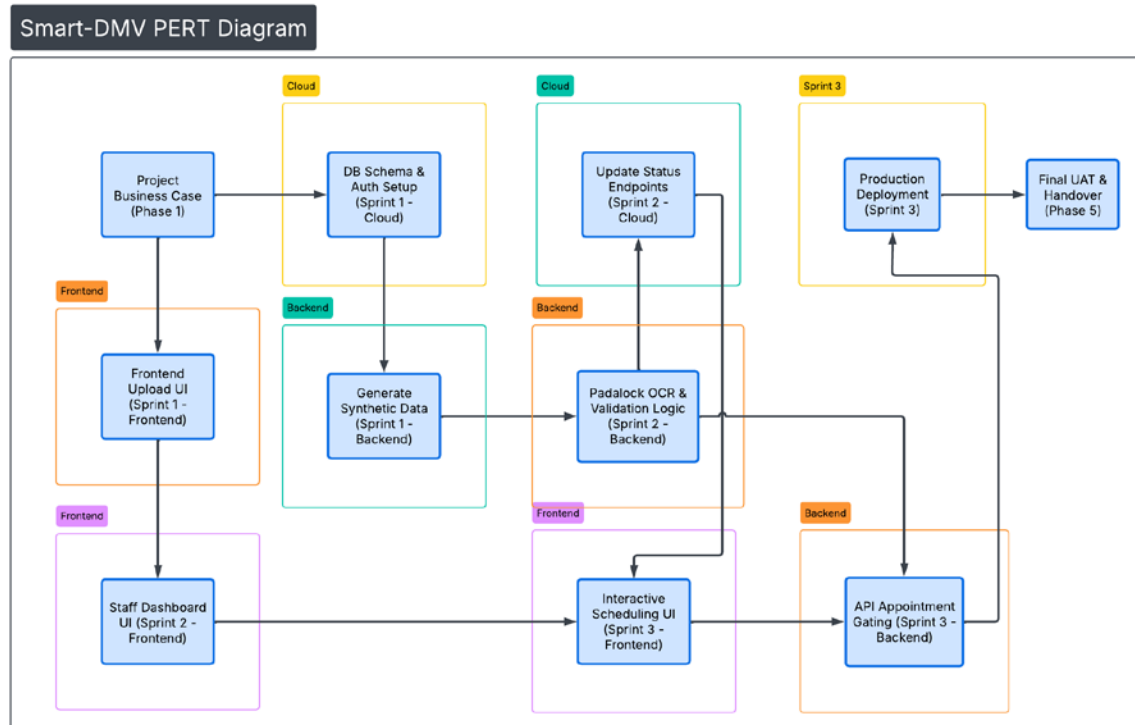
Smart DMV Activity Diagram



Note. Please refer to page no. 25

Figure 5*Smart DMV Use-Case Diagram*

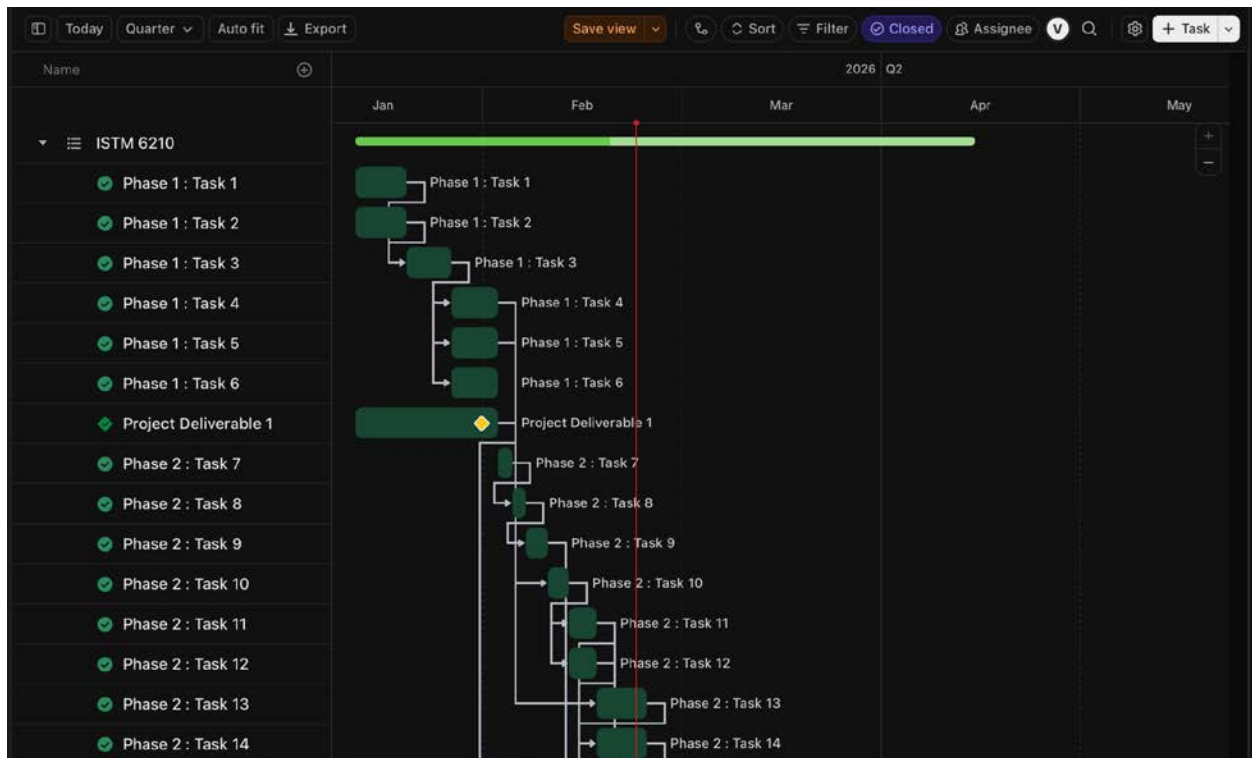
Note. Please refer to page no. 25

Figure 6*Smart DMV PERT Network Diagram*

Note. Please refer to page no. 37

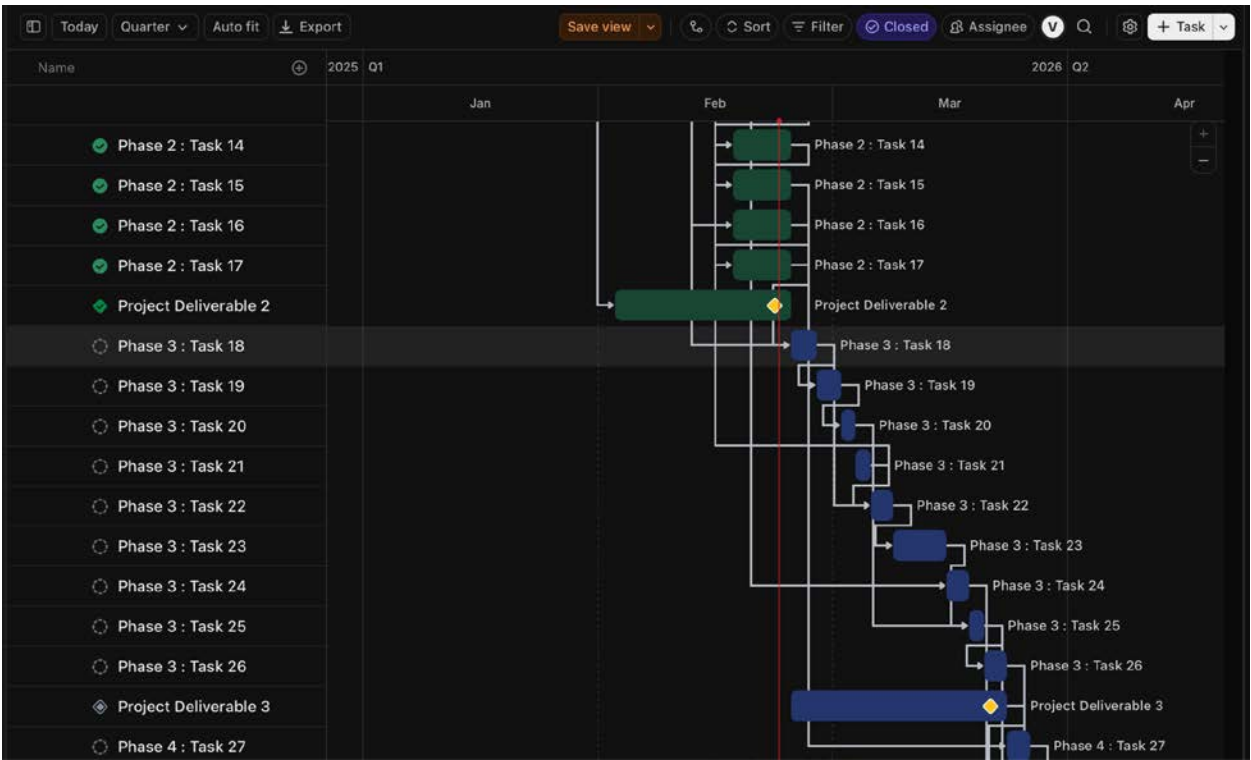
Figure 7

Timeline and milestone with Gantt Chart



Note. Please refer to page no. 39

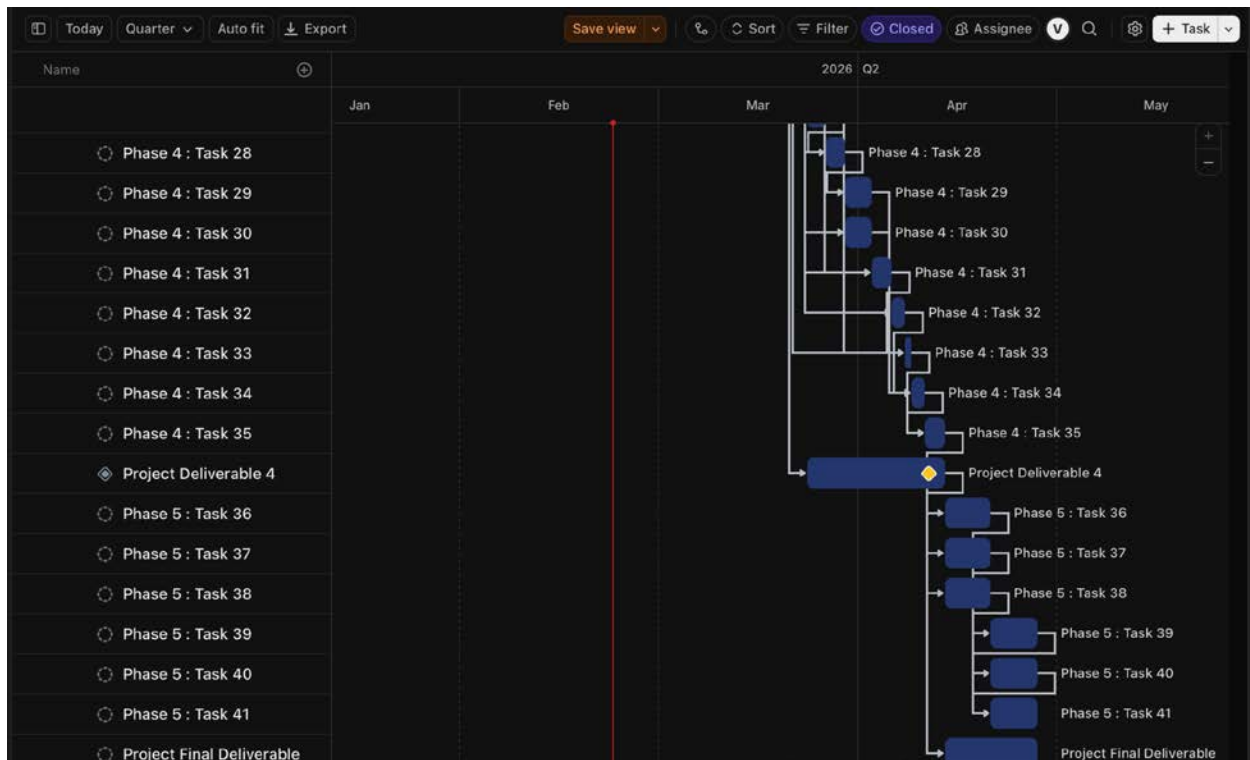
Figure 8
Timeline and milestone with Gantt Chart



Note. Please refer to page no. 39

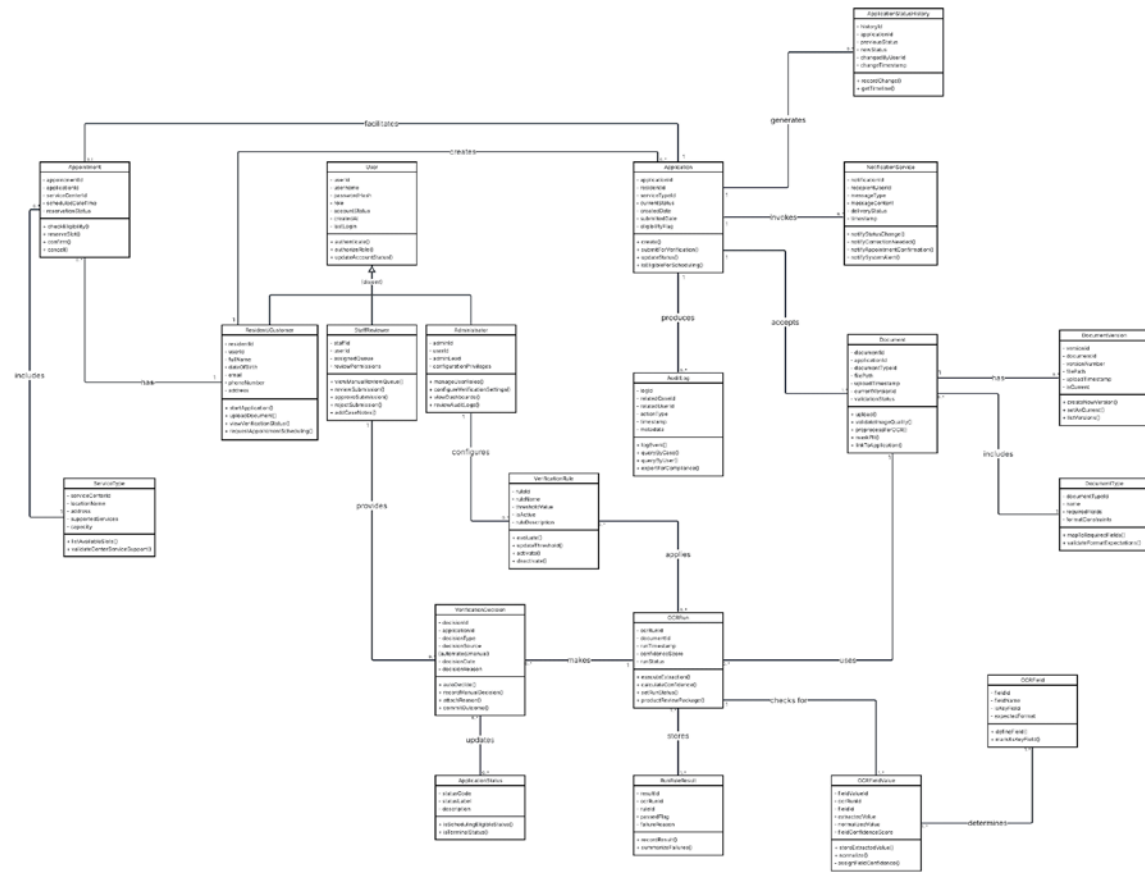
Figure 9

Timeline and milestone with Gantt Chart

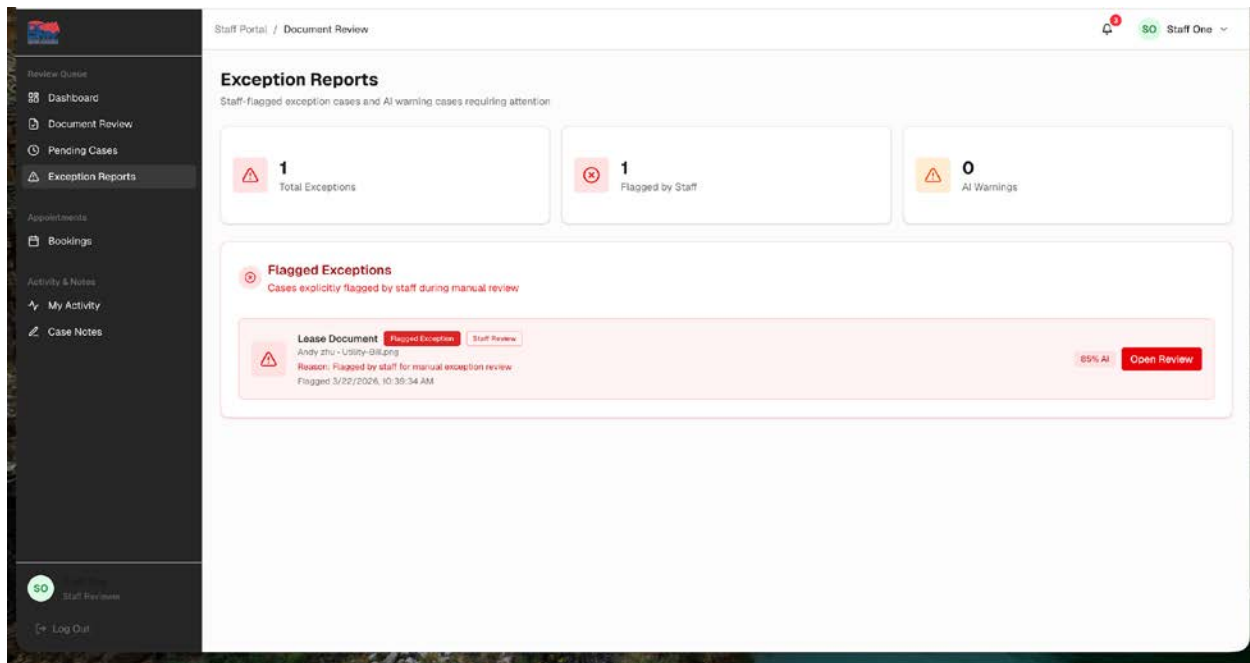


Note. Please refer to page no. 39

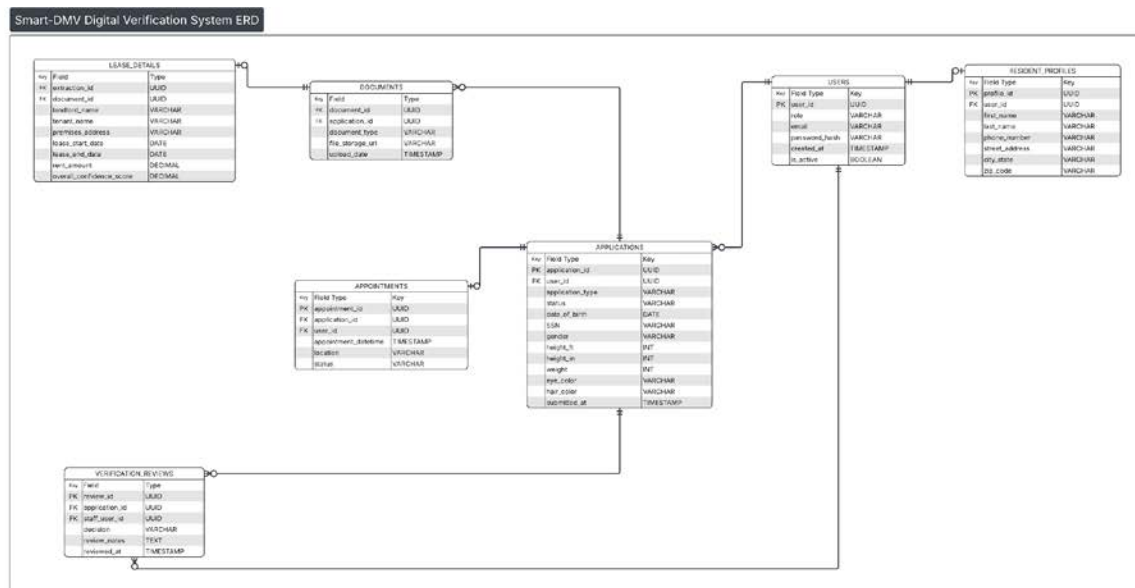
Class Diagram



Note. Please refer to page no. 47

Figure 11*Exception Reports*

Note. Please refer to page no. 47

Figure 12*DMV Entity- Relation Diagram Schema*

Note. Please refer to page no. 57

Figure 13*Image Quality Gatekeeper*

DMV Lease Verification System

Phase 1: Image Upload → Quality Check

Upload a lease document image

 Drag and drop file here
Limit: 200MB per file • JPG, JPEG, PNG, TIFF, TIF

Browse files

 lease_APP-0998_p11.jpg 407.6KB ×

Original Image

42. **ADDENDUM ATTACHED:** Yes ☒ No ☐

IN WITNESS WHEREOF, the parties hereto agree to abide by all of the terms and conditions in this lease agreement:

Tenant	<u>Rodriguez Vazquez</u>	Date	<u>2/17/2023</u>	Landlord	<u>Theresa Rodriguez</u>	Date	<u>2/17/2023</u>
Tenant		Date		Landlord		Date	
Tenant		Date		REALTOR® Firm			
Tenant		Date		Agent		Date	
Security Deposit Received:	\$		From		On	Date	
First Month's Rent Received:	\$		From		On	Date	
Pro-Rata Rent Received:	\$		From		On	Date	

signing, are agreeing to be guarantors of the prompt and faithful performance of all of the obligations of Tenant under the Lease. Guarantors consent to the jurisdiction of the courts of the District of Columbia in any action arising under the Lease or this Guaranty and agree to service of process in the manner prescribed in the so-called Long Arm statute of the District of Columbia as from time to time amended or superseded. Guarantors have no right of occupancy under this Lease. All signatures must be notarized unless witnessed by Landlord/Agent.

Name of Guarantor		Name of Guarantor	
Address of Guarantor		Address of Guarantor	
Signature of Guarantor	Date	Signature of Guarantor	Date
Name of Guarantor		Name of Guarantor	
Address of Guarantor		Address of Guarantor	
Signature of Guarantor	Date	Signature of Guarantor	Date
Name of Guarantor		Name of Guarantor	
Address of Guarantor		Address of Guarantor	
Signature of Guarantor	Date	Signature of Guarantor	Date

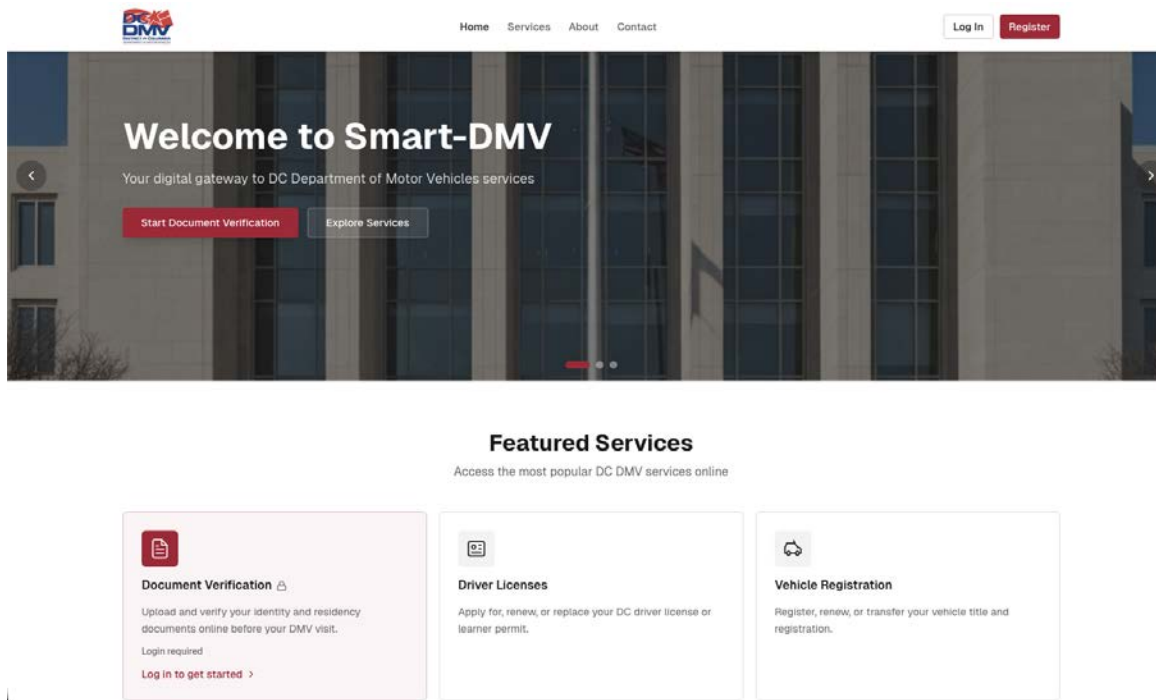
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This Recommended Form is the property of The Greater Capital Area Association of REALTORS®, Inc. and is for use by members only.
Previous editions of this form should be destroyed.
GCAAR Form # 1221 - DC Residential Lease Page 11 of 11
Produced with pdfComply by eplapix 16376 Fifteen Mile Road, Fraser, Michigan 48036 www.eplapix.com 1/2019 123 EXAMPLE

Quality Check Results

All quality checks passed!

Format	File Size	Resolution	Blur Score
Pass	Pass	1340x1724	2298.15

Note. Please refer to page no. 57

Figure 14*DMV Landing Page*

Note. Please refer to page no. 58

Figure 15*DMV Registration Page*

The image shows a screenshot of the DMV Registration Page. A modal window titled "Create an Account" is centered on the screen. The modal contains the following fields and text:

- Create an Account** (Title)
- Register to start your document verification process. Your information will pre-fill the application form.
- First Name**: John
- Last Name**: Doe
- Email**: you@example.com
- Phone Number**: (202) 555-0123
- Street Address**: 1234 Pennsylvania Ave NW
- City & State**: Washington, DC
- ZIP Code**: 20001
- Password**: At least 6 characters
- Confirm Password**: Confirm password
- Create Account** (Button)
- Already have an account? [Log In](#)

The background of the page is a blurred view of the DMV website, showing navigation links (Home, Services, About, Contact), a "Log In" button, and a "Register" button. There are also some news snippets visible in the background.

Note. Please refer to page no. 58

Figure 16*DMV Multiple Step Application Form*

The screenshot shows the DC DMV website's application form. At the top, there's a navigation bar with 'Home', 'Services', 'About', and 'Contact'. A user profile 'EE Enoch' is visible. The main heading is 'DC DRIVER LICENSE OR IDENTIFICATION CARD APPLICATION'. Below this, a progress bar shows four steps: 1. Personal Info (active), 2. History, 3. Preferences, and 4. Registration. A note states: 'The information you provide will be used to register you to vote unless you decline in Section G. Step 1 of 4: Application type & personal details.'

A. What do you need?

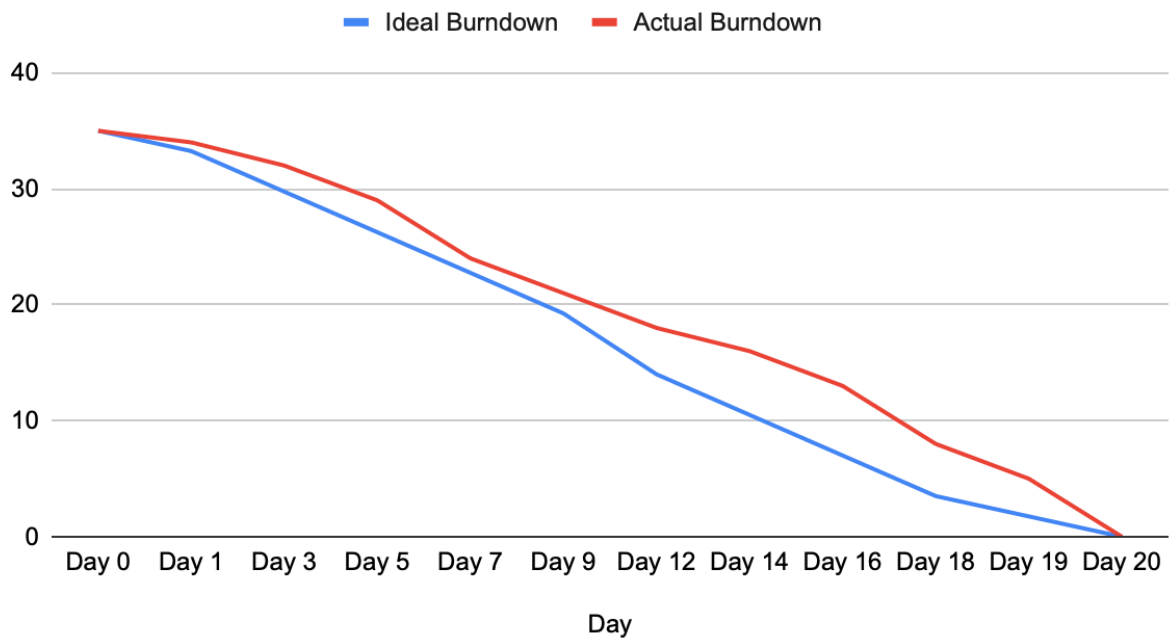
☐ Driver License ☐ Identification Card ☐ Motorcycle Endorsement

B. Tell us about yourself

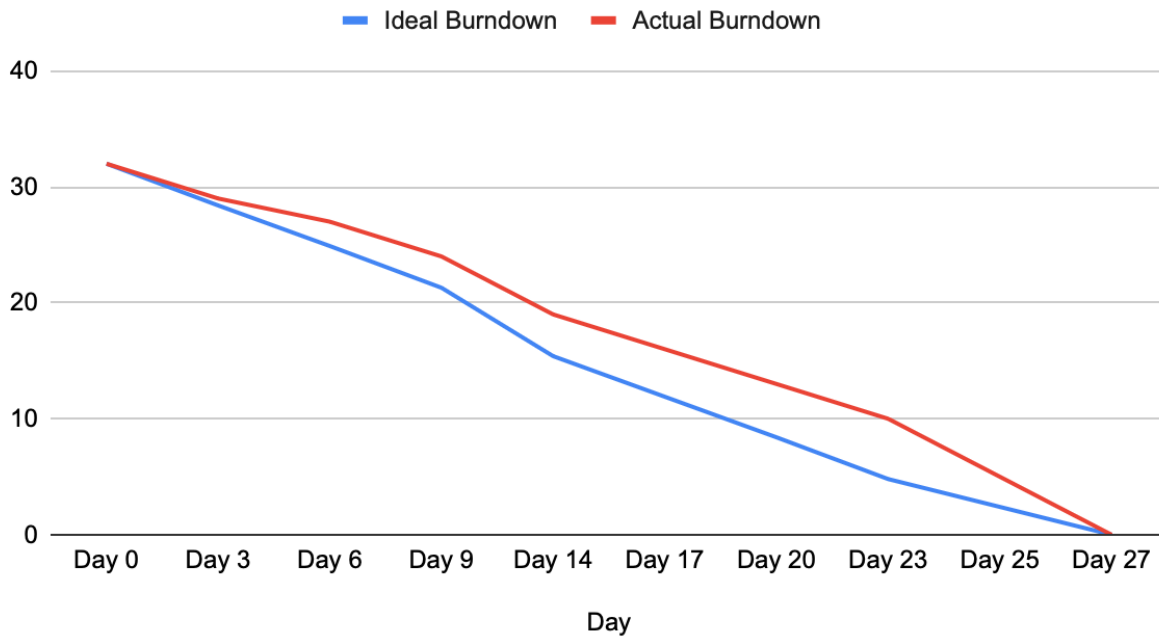
LAST NAME Saka		FIRST NAME Enoch Eugene	MIDDLE NAME JR./SR./III, ETC.
ADDRESS WHERE YOU LIVE (A MAILING ONLY ADDRESS CANNOT BE USED) 4823 W Braddock RD		APT/UNIT #	CITY & STATE Washington, DC
ZIP CODE 2001			
DATE OF BIRTH MM / DD / YYYY	SOCIAL SECURITY #	US CITIZEN ☐ Yes ☐ No	GENDER ☐ Male ☐ Female ☐ Unspecified
WEIGHT LBS	HEIGHT FT IN	HAIR COLOR	EYE COLOR
OTHER NAMES YOU HAVE USED ON A DRIVER LICENSE OR ID CARD			
CELL PHONE (703) 5825230	ALTERNATE PHONE ()	TEXT NOTIFICATION ☐ Yes Standard rates apply.	EMAIL s.eugene@gmail.gwu.edu

Navigation buttons at the bottom: '< Previous', 'Save Draft', and 'Next >'.

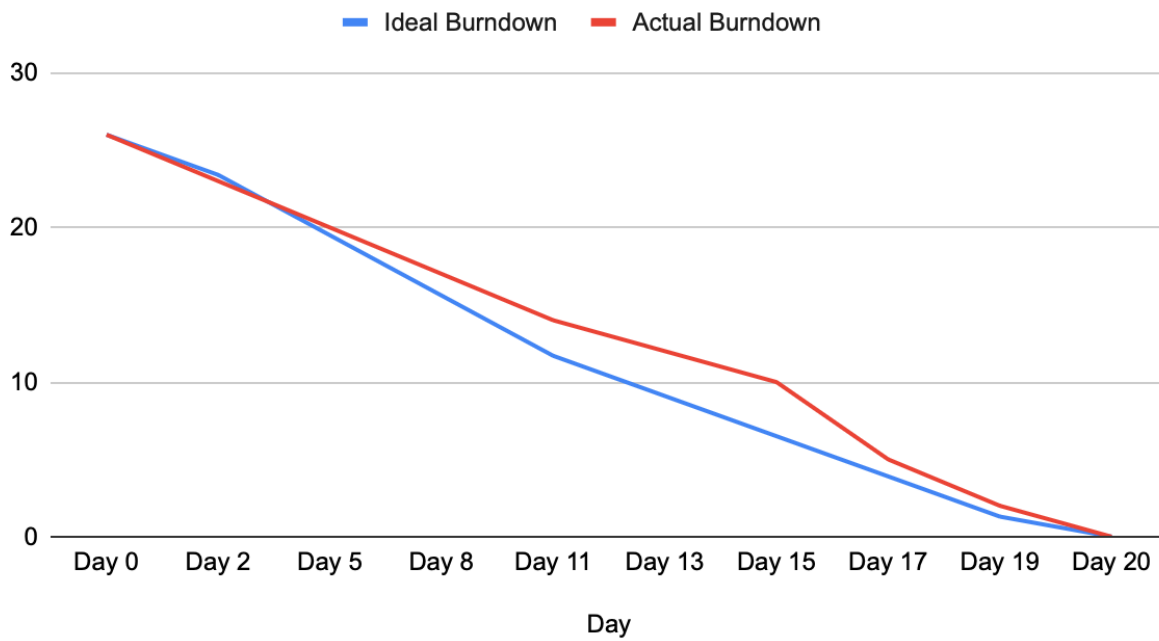
Note. Please refer to page no. 59

Figure 17*Sprint 1 Burndown Chart*

Note. Please refer to page no. 59

Figure 18*Sprint 2 Burndown Chart*

Note. Please refer to page no. 60

Figure 19*Sprint 3 Burndown Chart*

Note. Please refer to page no. 61